COMP2711 / DISCRETE MATHEMATICAL TOOLS

Induction and Recursion

Past Paper Solutions

76	2	2013-25
verified exercises	topic groups	source papers

English solutions aligned with the supplied official answers.

Each answer has two parts: the original Official Answer and the existing Supplementary Explanation. Source PDF pages are retained for verification.

Generated 2 October 2026

All verified, in-scope questions linked to Induction and Recursion in the knowledge base.

Each question is included once, including cross-topic applications.

READING GUIDE

Contents

Induction
29 exercises — I001-I029

Recursion
47 exercises — I030-I076

How to use this booklet

Questions are grouped by their first Induction and Recursion subtopic, then ordered by difficulty and year (newest first). Each question appears once as I001-I076. Difficulty ranges from 1 (introductory) to 4 (challenging).

Official Answer reproduces the answer in the source paper, with its physical PDF page reference.

Supplementary Explanation retains the existing independently authored reasoning, including the stated result and derivation. The two parts remain separate throughout the booklet.

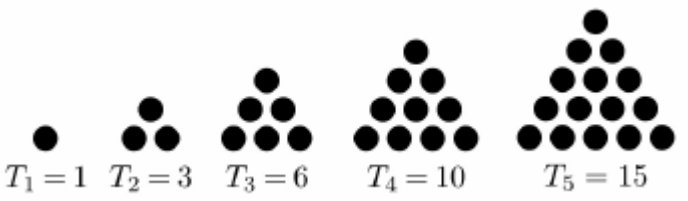
Source references use physical PDF pages. File links work with the local source collection at its existing location.

The answer provenance shown for each exercise retains its knowledge-base classification; attaching an official quotation does not reclassify a derived correction as official.

I001

Induction

2024 Fall Final | Question 9(b)(iii) | Source page 24 | Difficulty 2 | 2 marks
Q-2024F-FINAL-P09-b-iii | Answer provenance: official

> Consider the following picture:
>
> 
>
>
> Consider the sequence of triangles formed by circles. Let $T(n)$ be the number of circles needed to form a triangle with n circles on each side.
>
> (iii) Prove the closed formula for $T(n)$ using induction.

Official Answer

Source: physical PDF page 24 (COMP2711_F2024_Final.pdf).

> (iii) Base Case: $n = 1 : 1 = \frac{1 \cdot 2}{2}$. Induction hypothesis: $n = k$ then $T_k = \frac{k \cdot (k+1)}{2}$. For $n = k + 1$ we need to prove $T_{k+1} = \frac{(k+1) \cdot (k+2)}{2}$.
>
> $$T_{k+1} = T(k) + (k+1) = \frac{k \cdot (k+1)}{2} + \frac{2(k+1)}{2} = \frac{(k+1)(k+2)}{2}.$$

Supplementary Explanation

Answer. For all positive integers n, $T(n) = n(n+1)/2$.

Derivation/proof. Base case: $n = 1$ gives $T(1) = 1 = 1 \cdot 2/2$. The inductive hypothesis is $T(k) = k(k+1)/2$ for some $k \geq 1$. Using the established recurrence, $T(k+1) = T(k) + (k+1) = k(k+1)/2 + 2(k+1)/2 = (k+1)(k+2)/2$. This is precisely the formula with $n = k + 1$. The step and base case therefore establish the result for all positive integers n.

I002

Induction

2024 Spring Final | Question 1 | Source page 1 | Difficulty 2 | 3 marks
Q-2024S-FINAL-P01-N | Answer provenance: official

> Find an integer N such that $2^n > 3n^2$ whenever n is greater than N.

Official Answer

Source: physical PDF page 1 (COMP2711_S2024_Final.pdf).

> Answer: We want to find an integer N such that $2^n > 3n^2$ for all $n > N$. We choose $N = 7$. We will use induction to prove that $2^n > 3n^2$ whenever $n \geq 8$. Base case: For $n = 8$, we have $2^8 > 3(8)^2$, because $2^8 = 256$ and $3(8)^2 = 3 \cdot 64 = 192$. Induction step: For the inductive hypothesis, suppose that $2^k > 3k^2$ holds for some integer $k \geq 8$. We want to show that $2^{k+1} > 3(k+1)^2$. Applying the inductive hypothesis, we get $2^{k+1} = 2 \cdot 2^k > 2 \cdot 3k^2$. Thus
>
> $$2^{k+1} > 2(3k^2) = 3(2k^2) = 3(k^2 + k^2) \geq 3(k^2 + 3k),$$
>
> where in the last step we have used the fact that $k \geq 3$ (recall that $k \geq 8$). Thus
>
> $$2^{k+1} \geq 3(k^2 + 3k) = 3(k^2 + 2k + k) \geq 3(k^2 + 2k + 1),$$
>
> since $k \geq 1$ (recall that $k \geq 8$). Thus $2^{k+1} > 3(k+1)^2$, which completes the inductive step. Therefore, we have shown that $N = 7$ satisfies the condition that $2^n > 3n^2$ for all $n > N$.

Supplementary Explanation

Answer. Take $N = 7$.

Derivation/proof. $2^7 = 128 < 147 = 3 \cdot 7^2$, whereas $2^8 = 256 > 192 = 3 \cdot 8^2$, so choosing $N = 7$ makes $n = 8$ the starting case. For $k \geq 8$, $2k^2 - (k+1)^2 = k^2 - 2k - 1 > 0$. Thus, if $2^k > 3k^2$, then $2^{k+1} > 6k^2 > 3(k+1)^2$. Repeated application from the starting case covers every $n > 7$. Any larger integer threshold also works.

I003

Induction

2022 Final Part 1 | Question 4 | Source page 3 | Difficulty 2 | 8 marks
Q-2022-FINAL1-P04 | Answer provenance: official

> Use induction to prove that, for any integer $n \geq 1$, $5^n + 2 \cdot 11^n$ is divisible by 3.

Official Answer

Source: physical PDF page 3 (2022-final-part-1.pdf).

> Base case: When $n = 1$, we have $5^1 + 2 \cdot 11^1 = 27$, which is divisible by 3. So, the statement is true for $n = 1$.
>
> Induction hypothesis: Now let $n > 1$. Assume the statement is true for $n - 1$, i.e. $5^{n-1} + 2 \cdot 11^{n-1}$ is divisible by 3.
>
> Induction step: Consider the case of n:
>
> $$\begin{aligned} 5^n + 2 \cdot 11^n &= 5^{n-1} + 4 \cdot 5^{n-1} + 2 \cdot 11^{n-1} + 20 \cdot 11^{n-1} \\ &= 5^{n-1} + 2 \cdot 11^{n-1} + 4(5^{n-1} + 5 \cdot 11^{n-1}) \\ &= 5^{n-1} + 2 \cdot 11^{n-1} + 4(5^{n-1} + 2 \cdot 11^{n-1} + 3 \cdot 11^{n-1}) \\ &= 3y + 4(3y + 3 \cdot 11^{n-1}) \\ &= 3(5y + 4 \cdot 11^{n-1}) \end{aligned}$$
>
> which is divisible by 3.

Supplementary Explanation

Answer. $3 \mid (5^n + 2 \cdot 11^n)$ for all integers $n \geq 1$.

Derivation/proof. Base n=1:5+2·11=27, divisible by 3. Assume 5^k+2·11^k=3t for some integer t. Then $5^{k+1} + 2 \cdot 11^{k+1} = 5(5^k + 2 \cdot 11^k) + 12 \cdot 11^k = 15t + 12 \cdot 11^k = 3(5t + 4 \cdot 11^k)$. The final parenthesis is an integer, proving the next instance. Ordinary induction completes the required divisibility proof.

The base is $27 = 3 \cdot 9$. The supplementary step is $5(5^k + 2 \cdot 11^k) + 12 \cdot 11^k = 3(5t + 4 \cdot 11^k)$ with an integer in parentheses, explicitly using the induction hypothesis.

I004

Induction

2021 Final Part 1 | Question 4(a) | Source page 3 | Difficulty 2 | 2 marks
Q-2021-FINAL1-P04-a | Answer provenance: official

> Suppose that $P(n)$ is a predicate. Under each of the following conditions, find the integers n such that $P(n)$ must be true:
>
> $P(0)$ is true; for all positive integers n, if $P(n)$ is true, then $P(n+2)$ is true.

Official Answer

Source: physical PDF page 3 (2021-final-part-1.pdf).

> (a) 0 only.

Supplementary Explanation

Answer. Only $n = 0$.

Derivation/proof. The only supplied true instance is P(0), but the rule only permits positive n. Therefore it cannot start at zero. To prove no other integer is forced, define P to be true only at zero. Every implication with positive n then has a false antecedent and is true, so this predicate satisfies all assumptions while every other instance is false.

The explicit countermodel is $P(n) \iff n = 0$: zero is not an allowed starting value for the rule, and every other integer can be false.

I005

Induction

2021 Final Part 1 | Question 4(b) | Source page 3 | Difficulty 2 | 2 marks
Q-2021-FINAL1-P04-b | Answer provenance: official

> Suppose that $P(n)$ is a predicate. Under each of the following conditions, find the integers n such that $P(n)$ must be true:
>
> $P(4)$ is true; for all positive integers n, if $P(n)$ is true, then $P(2n)$ is true.

Official Answer

Source: physical PDF page 3 (2021-final-part-1.pdf).

> (b) For all integers n that is a power of 2 and greater than or equal to 4.

Supplementary Explanation

Answer. $n = 2^k$ for integers $k \geq 2$.

Derivation/proof. Starting from 4, repeated rule applications yield 4,8,16,..., namely 4·2^j for j≥0. Conversely let P hold exactly on this set. Doubling any member stays in the set; at any positive nonmember the implication has false antecedent. This predicate satisfies the assumptions and shows no other integers are forced.

The chain is $n = 4 \cdot 2^j = 2^{j+2}$ for integers $j \geq 0$. It includes neither 2, 1, 0, nor negative integers.

For the predicate true exactly on the chain, outside the chain the antecedent is false.

I006

Induction

2021 Final Part 1 | Question 4(c) | Source page 3 | Difficulty 2 | 2 marks
Q-2021-FINAL1-P04-c | Answer provenance: official

> Suppose that $P(n)$ is a predicate. Under each of the following conditions, find the integers n such that $P(n)$ must be true:
>
> $P(1)$ is true; for all positive integers n, if $P(n)$ and $P(n+1)$ are true, then $P(n+2)$ is true.

Official Answer

Source: physical PDF page 3 (2021-final-part-1.pdf).

> (c) 1 only.

Supplementary Explanation

Answer. Only $n = 1$.

Derivation/proof. To infer P(3), the first possible rule instance requires both P(1) and P(2), but P(2) is absent. No other step has two established consecutive premises either. Define P true only at 1: then every antecedent P(n)∧P(n+1) is false, so all implications hold. This witnesses that every integer except 1 can remain false.

Take $P(n) \iff n = 1$. Then every $P(n) \wedge P(n+1)$ is false, so no other integer is forced.

I007

Induction

2021 Final Part 1 | Question 4(d) | Source page 3 | Difficulty 2 | 2 marks
Q-2021-FINAL1-P04-d | Answer provenance: official

> Suppose that $P(n)$ is a predicate. Under each of the following conditions, find the integers n such that $P(n)$ must be true:
>
> $P(2)$ is true; for all positive integers n, if $P(k)$ is true for $2 \leq k \leq n$, then $P(n+1)$ is true.

Official Answer

Source: physical PDF page 3 (2021-final-part-1.pdf).

> (d) For all integers $n \geq 2$.

Supplementary Explanation

Answer. All integers $n \geq 2$.

Derivation/proof. P(2) is the base. If P(2),...,P(n) hold for n≥2, the rule gives P(n+1), so strong induction proves every integer at least 2. No smaller integer is forced: the predicate true exactly for integers ≥2 satisfies the rule. At n=1 the premise interval is empty and the conclusion is P(2), already included; it does not imply P(1).

The induction uses $P(2), \ldots, P(n)$. At $n = 1$, the range $2 \leq k \leq 1$ is empty, and the vacuous premise only yields $P(2)$.

I008

Induction

2016 Final | Question 7(c) | Source page 13 | Difficulty 2 | 3 marks
Q-2016-FINAL-P07-c | Answer provenance: official

> Assume that a party of n people wants to select some of their members to form a committee. The committee must have at least one member. Let $T(n)$ be the total number of possible committees that can be formed from a party of n people.
>
> (c) Prove that the closed form solution in part (b) is correct by induction.

Official Answer

Source: PDF physical page 13 (2016-final.pdf); PDF physical page 14 (2016-final.pdf).

(c) Base case ($n = 1$): $T(1) = 1 = 2^1 - 1$, so the base case is true.

Assume that $T(n-1) = 2^{n-1} - 1$ is true for $n > 1$,

$$\begin{aligned} T(n) &= 2T(n-1) + 1 \\ &= 2(2^{n-1} - 1) + 1 \\ &= 2^n - 2 + 1 = 2^n - 1 \end{aligned}$$

By the principle of mathematical induction, we have proved that $T(n) = 2^n - 1$ is true for $n \geq 1$.

Supplementary Explanation

Answer. For every $n \geq 1$, $T(n) = 2^n - 1$.

Derivation/proof. Base case: $T(1) = 1 = 2^1 - 1$. Assume $T(n-1) = 2^{n-1} - 1$ for $n > 1$. Then $T(n) = 2T(n-1) + 1 = 2(2^{n-1} - 1) + 1 = 2^n - 1$. By induction the formula holds for all positive integers.

The constant simplification is $2^n - 2 + 1 = 2^n - 1$. This induction verifies the closed form, whereas part (a) justified the combinatorial recurrence.

I009

Induction

2015 Final | Question 8(b) | Source page 14 | Difficulty 2 | 6 marks
Q-2015-FINAL-P08-b | Answer provenance: derived

(a) Consider the recurrence below defined for $n \geq 0$.

$$T(n) = \begin{cases} 3 & \text{if } n = 0, \\ 7T(n-1) + 12 & \text{if } n > 0. \end{cases}$$

(b) Prove the correctness of your solution by induction.

Official Answer

Source: PDF physical page 14 (2015-final.pdf).

(b) We need to prove:

$$T(n) = 5 \cdot 7^n - 2 \qquad (1)$$

Base case: When $n = 0$, $T(0) = 3 = 5 \cdot 7^0 - 2$. Equation (1) is true.

Inductive step: Now consider $n > 1$. Assume Equation (1) is true for the case of $n - 1$, i.e.,

$$T(n-1) = 5 \cdot 7^{n-1} - 2.$$

For the case of n, we have

$$\begin{aligned} T(n) &= 7T(n-1) + 12 \\ &= 7(5 \cdot 7^{n-1} - 2) + 12 \quad \text{(induction hypothesis)} \\ &= 5 \cdot 7^n - 2. \end{aligned}$$

By the weak principle of mathematical induction, we conclude that Equation (1) is true for all $n \geq 0$.

Supplementary Explanation

Answer. $T(n) = 5 \cdot 7^n - 2$ for every integer $n \geq 0$.

Derivation/proof. Base: $T(0) = 3 = 5 \cdot 7^0 - 2$. Let $n \geq 1$ and assume $T(n-1) = 5 \cdot 7^{n-1} - 2$. Then $T(n) = 7(5 \cdot 7^{n-1} - 2) + 12 = 5 \cdot 7^n - 2$. Induction proves the formula. The printed proof says $n > 1$ in the step; using $n \geq 1$ includes the otherwise omitted transition from 0 to 1.

The constant terms combine as $-14 + 12 = -2$.

Expanding the inductive substitution gives $5 \cdot 7^n - 14 + 12 = 5 \cdot 7^n - 2$.

I010

Induction

2025 Fall Final | Question 9(b) | Source page 14 | Difficulty 3 | 5 marks
Q-2025F-FINAL-P09-b | Answer provenance: official

> (b) [5 points] Prove your result in part (a) using induction.

Official Answer

Source: physical PDF page 15 (COMP2711_F2025_Final.pdf), physical PDF page 16 (COMP2711_F2025_Final.pdf).

(b) Let $c = \log_4 3$. We prove that for all $n = 4^k$ ($k \geq 0$),

$$T(n) = \frac{16n^2 - 3n^c}{13}.$$

Base case: $n = 4^0 = 1$

$$T(1) = 1, \qquad \frac{16 \cdot 1^2 - 3 \cdot 1^c}{13} = \frac{16 - 3}{13} = \frac{13}{13} = 1.$$

Holds.

Inductive hypothesis: Assume that for some $m = 4^\ell$ ($\ell \geq 0$),

$$T(m) = \frac{16m^2 - 3m^c}{13}.$$

Inductive step: Consider $n = 4^{\ell+1} = 4m$. By the recurrence,

$$T(n) = 3T\left(\frac{n}{4}\right) + n^2 = 3T(m) + n^2.$$

Substitute the inductive hypothesis and $n = 4m$ (so $n^2 = 16m^2$):

$$\begin{aligned} T(n) &= 3 \cdot \frac{16m^2 - 3m^c}{13} + 16m^2 \\ &= \frac{48m^2 - 9m^c}{13} + 16m^2 \\ &= \frac{48m^2 - 9m^c + 208m^2}{13} \\ &= \frac{256m^2 - 9m^c}{13}. \end{aligned}$$

Now note that $n^c = (4m)^c = 4^c m^c = (4^{\log_4 3})m^c = 3m^c$, so $9m^c = 3 \cdot 3m^c = 3n^c$. Also $256m^2 = 16 \cdot 16m^2 = 16n^2$. Therefore

$$T(n) = \frac{16n^2 - -3n^c}{13},$$

which is exactly the formula for n. By the principle of mathematical induction, the closed-form holds for all $n = 4^k$, $k \geq 0$.

Supplementary Explanation

Answer. For every $n = 4^k$ with $k \geq 0$, $T(n) = (16n^2 - 3n^{\log_4 3})/13$.

Derivation/proof. Use the recurrence $T(n) = 3T(n/4) + n^2$, the initial value $T(1) = 1$, and the candidate from part (a). Put $c = \log_4 3$. At $n = 1$, $(16 - 3)/13 = 1 = T(1)$. Assume $T(m) = (16m^2 - 3m^c)/13$ for $m = 4^k$. For $n = 4m$, substitution gives

$$T(4m) = 3(16m^2 - 3m^c)/13 + 16m^2 = (256m^2 - 9m^c)/13.$$

Since $16n^2 = 256m^2$ and $n^c = (4m)^c = 4^c m^c = 3m^c$, this equals $(16n^2 - 3n^c)/13$. The induction is complete. Keeping a single minus sign is essential; interpreting the source's double minus literally would already fail the base value.

I011

Induction

2025 Fall Final | Question 10(iii) | Source page 17 | Difficulty 3 | 4 marks
Q-2025F-FINAL-P10-iii | Answer provenance: official

> A spider starts with 10 legs. Each day, its legs instantly double by splitting, but immediately after, exactly 2 legs fall off and are lost forever. Let a_n be the number of legs after n days, with $a_0 = 10$.
>
> (iii) [4 points] Prove that your closed-form formula is correct using mathematical induction.

Official Answer

Source: physical PDF page 18 (COMP2711_F2025_Final.pdf).

(iii) Proof by mathematical induction.

Base case ($n = 0$):

$$8 \cdot 2^0 + 2 = 8 \cdot 1 + 2 = 10 = a_0 \quad \checkmark$$

Inductive hypothesis: Assume that for some integer $k \geq 0$, $a_k = 8 \cdot 2^k + 2$.

Inductive step: Using the recurrence,

$$\begin{aligned} a_{k+1} &= 2a_k - 2 && (1)\\ &= 2(8 \cdot 2^k + 2) - 2 && (2)\\ &= 16 \cdot 2^k + 4 - 2 && (3)\\ &= 16 \cdot 2^k + 2 && (4)\\ &= 8 \cdot 2^{k+1} + 2. && (5) \end{aligned}$$

This is exactly the formula for $n = k + 1$.

By mathematical induction, $a_n = 8 \cdot 2^n + 2$ holds for all integers $n \geq 0$.

Supplementary Explanation

Answer. $a_n = 8 \cdot 2^n + 2$ for every integer $n \geq 0$.

Derivation/proof. Use $a_0 = 10$ and $a_{n+1} = 2a_n - 2$. The base case is $8 \cdot 2^0 + 2 = 10 = a_0$. Assume $a_k = 8 \cdot 2^k + 2$ for some $k \geq 0$. By the recurrence,

$$a_{k+1} = 2a_k - 2 = 2(8 \cdot 2^k + 2) - 2 = 16 \cdot 2^k + 2 = 8 \cdot 2^{k+1} + 2.$$

This is precisely the proposed expression for day $k + 1$. Mathematical induction therefore establishes the formula for every nonnegative integer day.

I012

Induction

2024 Spring Final | Question 1 | Source page 1 | Difficulty 3
Q-2024S-FINAL-P01-proof | Answer provenance: official

> Prove that your result is correct using mathematical induction.

Official Answer

Source: physical PDF page 1 (COMP2711_S2024_Final.pdf).

> Answer: We want to find an integer N such that $2^n > 3n^2$ for all $n > N$. We choose $N = 7$. We will use induction to prove that $2^n > 3n^2$ whenever $n \geq 8$. Base case: For $n = 8$, we have $2^8 > 3(8)^2$, because $2^8 = 256$ and $3(8)^2 = 3 \cdot 64 = 192$. Induction step: For the inductive hypothesis, suppose that $2^k > 3k^2$ holds for some integer $k \geq 8$. We want to show that $2^{k+1} > 3(k+1)^2$. Applying the inductive hypothesis, we get $2^{k+1} = 2 \cdot 2^k > 2 \cdot 3k^2$. Thus
>
> $$2^{k+1} > 2(3k^2) = 3(2k^2) = 3(k^2 + k^2) \geq 3(k^2 + 3k),$$
>
> where in the last step we have used the fact that $k \geq 3$ (recall that $k \geq 8$). Thus
>
> $$2^{k+1} \geq 3(k^2 + 3k) = 3(k^2 + 2k + k) \geq 3(k^2 + 2k + 1),$$
>
> since $k \geq 1$ (recall that $k \geq 8$). Thus $2^{k+1} > 3(k+1)^2$, which completes the inductive step. Therefore, we have shown that $N = 7$ satisfies the condition that $2^n > 3n^2$ for all $n > N$.

Supplementary Explanation

Answer. $2^n > 3n^2$ for every integer $n \geq 8$.

Derivation/proof. Use the preceding choice $N = 7$. Base case: at $n = 8$, $2^8 = 256 > 192 = 3 \cdot 8^2$. Inductive hypothesis: suppose $2^k > 3k^2$ for some $k \geq 8$. Then $2^{k+1} = 2 \cdot 2^k > 6k^2$. Since $k \geq 8$, $2k^2 - (k+1)^2 = (k-1)^2 - 2 \geq 49 - 2 > 0$, so $6k^2 > 3(k+1)^2$. Combining these strict inequalities proves the next case. Mathematical induction gives the result for all $n \geq 8$, equivalently $n > N$.

I013

Induction

2024 Spring Final | Question 9(b) | Source page 11 | Difficulty 3 | 10 marks
Q-2024S-FINAL-P09-b | Answer provenance: official

(b) Let p be a prime number. Use mathematical induction to prove that for any positive integer a, the number $a^p - a$ is divisible by p. In other words, you need to prove Fermat's Little Theorem using induction on the integer a. Note: If you give the number-theoretic proof of Fermat's Little Theorem presented in class, you will receive 0 marks. Hint: In your proof of part (b), you may find the claim given in part (a) useful. Note that in part (b), you are permitted to use this claim from part (a), even if you were unable to prove it.

Official Answer

Source: physical PDF page 11 (COMP2711_S2024_Final.pdf), physical PDF page 12 (COMP2711_S2024_Final.pdf).

(b) The base case is when $a = 1$, in which case we have $a^p - a = 1^p - 1 = 0$, which is divisible by p, and hence the base case holds. Now supposing that the claim holds for an arbitrary a, i.e., $p \mid a^p - a$, we wish to show that it holds for $a + 1$, i.e., $p \mid (a+1)^p - (a+1)$. Expanding $(a+1)^p - (a+1)$ using the binomial formula, we get

$$\begin{aligned}
(a+1)^p - (a+1) &= \left(\sum_{i=0}^{p} \binom{p}{i} \cdot a^i \cdot 1^{p-i}\right) - (a+1) \\
&= \left(\sum_{i=0}^{p} \binom{p}{i} \cdot a^i\right) - (a+1) \\
&= \left(\sum_{i=1}^{p-1} \binom{p}{i} \cdot a^i\right) + \binom{p}{0} \cdot a^0 + \binom{p}{p} \cdot a^p - (a+1) \\
&= \left(\sum_{i=1}^{p-1} \binom{p}{i} \cdot a^i\right) + 1 + a^p - (a+1) \\
&= \left(\sum_{i=1}^{p-1} \underbrace{\binom{p}{i}}_{\text{divisible by } p \text{ from part (a)}} \cdot a^i\right) + \underbrace{a^p - a}_{\text{divisible by } p \text{ by IH}}.
\end{aligned}$$

Thus, $(a+1)^p - (a+1)$ is equal to the sum of terms each of which is divisible by p, and therefore we have $p \mid (a+1)^p - (a+1)$, which completes our inductive proof.

Supplementary Explanation

Answer. $p \mid a^p - a$ for every positive integer a.

Derivation/proof. Fix the prime p. Base case: $a = 1$ gives $1^p - 1 = 0$, divisible by p. The inductive hypothesis is $p \mid a^p - a$. By the binomial theorem, $(a+1)^p - (a+1) = (a^p - a) + \sum_{i=1}^{p-1} \binom{p}{i} a^i$: the constant one cancels the subtracted one, and $a^p - a$ forms the first term. Part (a) makes every internal coefficient, $1 \le i \le p-1$, divisible by p, so the entire sum is divisible by p. The first term is divisible by p by the inductive hypothesis. Thus $p \mid (a+1)^p - (a+1)$, completing induction without assuming Fermat's theorem.

I014

Induction

2020 Final Part 1 | Question 5(a) | Source page 3 | Difficulty 3
Q-2020-FINAL1-P05-a-proof | Answer provenance: official

> Consider the following series of functions defined recursively: $f_0(x) = x^{-1}$; for $n \geq 0$, $f_{n+1}(x) = f_n'(x)$ (recall that $(ax^b)' = abx^{b-1}$).
>
> Use induction to prove its correctness.

Official Answer

Source: physical PDF page 4 (2020-final-part-1.pdf).

$f_n(x) = (-1)^n n! \cdot x^{-(n+1)}$. We can prove this by induction. First $f_0(x) = (-1)^0 \cdot 0! \cdot x^{-1} = x^{-1}$. Suppose for some $k \geq 0$, $f_k(x) = (-1)^k k! \cdot x^{-(k+1)}$, then

$$f_{k+1}(x) = f\prime_k(x) = (-1)^k k! \cdot [-(k+1)] \cdot x^{-(k+2)} = (-1)^{k+1}(k+1)! \cdot x^{-(k+2)}.$$

Supplementary Explanation

Answer. The formula $f_n(x) = (-1)^n n! x^{-(n+1)}$ holds for every $n \geq 0$ and $x \neq 0$.

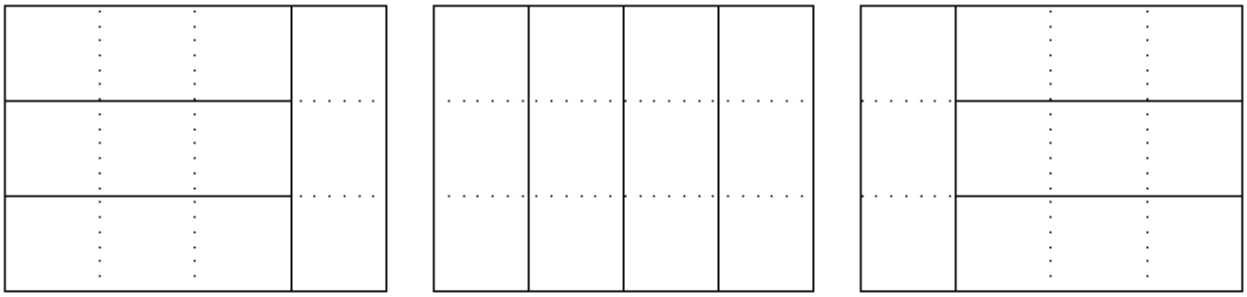
Derivation/proof. At n=0 the expression is $(-1)^0 0! x^{-1} = x^{-1} = f_0(x)$. Assume the formula holds at k. By the given recurrence and power rule, $f_{k+1}(x) = [(-1)^k k! x^{-(k+1)}]' = (-1)^k k! [-(k+1)] x^{-(k+2)} = (-1)^{k+1}(k+1)! x^{-(k+2)}$. This is exactly the asserted formula at k+1. Induction proves it on the inherited nonzero domain.

For $n = 0$, the proposed formula is $(-1)^0 0! x^{-1} = x^{-1} = f_0(x)$. Differentiating the induction hypothesis gives $(-1)^k k! [-(k+1)] x^{-(k+2)} = (-1)^{k+1}(k+1)! x^{-(k+2)}$, precisely the next formula for $x \neq 0$.

I015

Induction

2018 Final | Question 2(c) | Source page 4 | Difficulty 3
Q-2018-FINAL-P02-c | Answer provenance: official

> Let E_n be the number of ways in which you can tile a 3-by-n board with 1-by-3 pieces. For example, when $n = 4$ the board is 3-by-4 and $E_4 = 3$. The 3 ways are shown as follows:
>
> 
>
> (c) Let F_n be the Fibonacci numbers. Prove that $E_n \leq F_n$ for all $n \geq 1$ using induction. You may use the result of (b).

Official Answer

Source: physical PDF page 4 (2018-final.pdf).

We prove this statement $P(n)$ using induction on three base cases. Observe that $E_1 = F_1$ ($P(1)$), $E_2 = F_2$ ($P(2)$), and $E_3 = F_3$ ($P(3)$). For the inductive step, suppose that $P(k-1)$, $P(k-2)$, and $P(k-3)$ are true. Then $E_n = E_{n-1} + E_{n-3} \leq F_{n-1} + F_{n-3} \leq F_{n-1} + F_{n-2} = F_n$. (For the last inequality, we use the fact that the Fibonacci sequence is non-decreasing.) So $P(k)$ is true, thus proving the statement $P(n)$ by induction.

Supplementary Explanation

Answer. $E_n \leq F_n$ for every $n \geq 1$, with $F_1 = F_2 = 1$.

Derivation/proof. The base values are $E_1 = 1 = F_1$, $E_2 = 1 = F_2$, $E_3 = 2 = F_3$. For $n \geq 4$, assume the inequality for all smaller positive indices. Then $E_n = E_{n-1} + E_{n-3} \leq F_{n-1} + F_{n-3} \leq F_{n-1} + F_{n-2} = F_n$. The middle inequality follows because the positive Fibonacci sequence is nondecreasing. Strong induction completes the proof.

The required initial matches are $E_1 = F_1 = 1$, $E_2 = F_2 = 1$, and $E_3 = F_3 = 2$. Since the recurrences also agree, strong induction proves equality for every later index.

I016

Induction

2018 Final | Question 3(a) | Source page 5 | Difficulty 3
Q-2018-FINAL-P03-a | Answer provenance: official

> S, a set of integers, is recursively defined as follows:
>
> Basis: 4914 and 3588 are in S.
>
> Recursive step: If $x, y \in S$, then $x - y \in S$.
>
> (Note that x and y may be the same.)
>
> (a) Let $P(n)$ be the statement "for any integer a, if $a \in S$ then $an \in S$ and $-an \in S$." Prove $P(n)$ is true for all positive integers n by induction.

Official Answer

Source: physical PDF page 5 (2018-final.pdf).

$P(0)$ is true because if $a \in S$, applying the recursive step, $a - a = 0 \in S$. $P(1)$ is true since $a \in S$ and $0 - a = -a \in S$. Suppose $P(1)$ and $P(k)$ are true. Then $P(k+1)$ is true because, applying the recursive step, $-an - a = -(a+1)n \in S$ and $an - (-a) = (a+1)n \in S$.

Supplementary Explanation

Answer. Every positive and negative integer multiple of an element of S lies in S.

Derivation/proof. For any $a \in S$, closure gives $0 = a - a \in S$ and $-a = 0 - a \in S$, proving $P(1)$. Assume $ak, -ak \in S$. Then $a(k+1) = ak - (-a) \in S$ and $-a(k+1) = (-ak) - a \in S$. These are exactly the two assertions for $k+1$, so induction proves $P(n)$. Zero multiples were separately obtained at the start.

I017

Induction

2018 Final | Question 6(b) | Source page 8 | Difficulty 3
Q-2018-FINAL-P06-b | Answer provenance: official

> In this question, we consider simple graphs $G = (V, E)$ with $|V| \geq 3$ where all vertices satisfy $deg(v) \geq 2$. Let us call them two-dense graphs.
>
> Consider the following statement:
>
> All two-dense graphs have a triangle subgraph (a cycle of length 3).
>
> Here is an inductive proof of the statement.
>
> Proof. Let the proposition $P(n)$ be the above statement for $|V| = n$. We will prove $P(n)$ is true for $n \geq 3$.
>
> Base case ($P(3)$): For the two-dense graph G with three vertices, each vertex has a degree of 2, so G is simply a triangle and has itself as a subgraph.
>
> Inductive step: Suppose $P(k)$ is true, i.e. any two-dense graph with k vertices has a triangle subgraph. Extend any such graph G to obtain $G' = (V', E')$ by adding one vertex and at least two new edges from the new vertex to any different vertices in $|V|$. G' is a simple graph with $k + 1$ vertices and all vertices satisfy $deg(v) \geq 2$, so G' is two-dense. G' contains G as a subgraph, so using the inductive assumption, G' also has a triangle subgraph. Therefore, $P(k + 1)$ is true.
>
> (b) Since the statement is wrong, the proof must be wrong as well. Choose one of the following statements to explain why the proof is wrong, and explain it.
>
> 1. There is an error in the base case.
> 2. The proof requires another base case, which has not been proven.
> 3. In the inductive step, G' is wrongly constructed; sometimes it is not a two-dense graph.
> 4. In the inductive step, G' may not always have a triangle subgraph.
> 5. In the inductive step, G' is correct and always has a triangle subgraph, but that does not show $P(k + 1)$.

Official Answer

Source: physical PDF page 9 (2018-final.pdf).

The correct choice is (5). Not all graphs with $k + 1$ vertices satisfying $\deg(v) \geq 2$ can be constructed using the procedure in the inductive step. However, $P(k + 1)$ is a statement on all such graphs. Therefore, $P(k + 1)$ being true does not follow from G' containing a triangle.

Supplementary Explanation

Answer. Choice 5.

Derivation/proof. The extension really is two-dense and retains the old triangle. However $P(k + 1)$ quantifies over every two-dense graph with $k + 1$ vertices, not merely the outputs of that extension. For example, C_4 cannot arise by adding a vertex and edges to a triangle, because such an operation never deletes the triangle. The induction step covers only a proper subclass and so cannot establish the universal statement.

The base case and the constructed graph are valid, so choices 1, 3, and 4 misidentify the issue. Adding another base case would not make the construction cover every graph either.

Choose 5, the option identifying the missing universal coverage of the claimed induction extension. The step only constructs a special family; it does not establish the property for all $k+1$-vertex graphs, so it cannot justify the universal claim.

I018

Induction

2017 Final | Question 8 | Source page 12 | Difficulty 3 | 6 marks
Q-2017-FINAL-P08 | Answer provenance: official

> Let $P(n)$ be the statement that a postage of n cents can be formed using just 4-cent and 7-cent stamps. Prove by induction that $P(n)$ is true for $n \geq 18$.

Official Answer

Source: physical PDF page 12 (2017-final.pdf).

By (weak) induction: It is easy to show that $P(18)$ is true. Now suppose that $P(k)$ is true where $k \geq 18$. We show that $P(k+1)$ is also true. Consider the postage of k cents (by inductive hypothesis, such postage can be formed). This postage either (i) contains at least one seven-cent stamp, or (ii) contains no seven-cent stamp. In this case, it contains at least five 4-cents stamp since $k \geq 18$. In case (i), by simply replace that seven-cent stamp by two 4-cet stamps, we obtain a postage of $k+1$ cents. In case (ii), by replacing that five 4-cent stamps by three 7-cent stamps, we obtain a postage of $k+1$ cents. This completes the inductive step. By strong induction: It is easy to show that $P(18), P(19), P(20), P(21)$ is true. Now suppose that $P(k-3), P(k-2), P(k-1), P(k)$ is true where $k \geq 21$. We show that $P(k+1)$ is true. Since $P(k-3)$ is true, a postage of $k-3$ cents can be formed. Adding one more 4-cent stamp, we obtain a postage of $k+1$ cents.

Supplementary Explanation

Answer. Every integer postage of at least 18 cents is possible.

Derivation/proof. Base cases: $18 = 7+7+4$, $19 = 7+4+4+4$, $20 = 5 \cdot 4$, and $21 = 3 \cdot 7$. Suppose all postages from 18 through k are possible, where $k \geq 21$. Then $k+1-4 = k-3 \geq 18$, so the induction hypothesis supplies postage $k-3$. Adding one 4-cent stamp makes $k+1$. Strong induction proves the result.

I019

Induction

2017 Final | Question 9 | Source page 13 | Difficulty 3 | 6 marks
Q-2017-FINAL-P09 | Answer provenance: official

> Use induction to prove that, for any non-negative integer n, $1 + rn \leq (1 + r)^n$, if $r > -1$.

Official Answer

Source: physical PDF page 13 (2017-final.pdf).

Base case: When $n = 0$,

$$\text{LHS} = 1 + r \cdot 0 = 1, \qquad \text{RHS} = (1 + r)^0 = 1.$$

LHS $\leq$ RHS. So, the statement is true for $n = 0$. Induction hypothesis: Now let $n \geq 1$. Assume the statement is true for $n - 1$, i.e.,

$$1 + r \cdot (n - 1) \leq (1 + r)^{n-1}.$$

Induction step: Consider the case of n,

$$\begin{aligned}
(1 + r)^n &= (1 + r) \cdot (1 + r)^{n-1} \\
&\geq (1 + r) \cdot (1 + r \cdot (n - 1)) && \text{by hypothesis} \\
&= 1 + rn + r^2 n - r^2 \\
&= 1 + rn + r^2(n - 1) \\
&\geq 1 + rn && \text{since } r^2(n - 1) \geq 0.
\end{aligned}$$

So, the statement is true for the case of n. By the principle of Mathematical Induction, we conclude that the statement is true for any non-negative integer n.

Supplementary Explanation

Answer. The inequality holds for all $n \geq 0$ and $r > -1$.

Derivation/proof. At $n = 0$, both sides equal 1. Assume $(1 + r)^k \geq 1 + rk$ for some $k \geq 0$. Since $1 + r > 0$, multiply to get $(1 + r)^{k+1} \geq (1 + r)(1 + rk) = 1 + r(k + 1) + kr^2 \geq 1 + r(k + 1)$. The last inequality uses $k \geq 0, r^2 \geq 0$. This proves the next case and hence all nonnegative integers.

I020

Induction

2015 Final | Question 10 | Source page 17 | Difficulty 3 | 10 marks
Q-2015-FINAL-P10 | Answer provenance: derived

> Prove the following statement by induction
>
> $$\big(n(n-1)\cdots(2)(1)+1\big) \bmod m = 1$$
>
> for all integer $n > 1$ and for any $m \in \{2, 3, \cdots, n\}$.
>
> Note: No marks will be awarded if you do not use the principle of induction.

Official Answer

Source: PDF physical page 17 (2015-final.pdf).

Let $p(n)$ denotes $(n(n-1)\cdots(2)(1)+1) \bmod m = 1$.

Base case ($n = 2$): $(2+1) \bmod 2 = 1$, so $p(2)$ is true.

Inductive hypothesis: Assume $p(n-1)$ is true for $n - 1 > 1$.

$$((n-1)\cdots(2)(1)+1) \bmod m = 1$$

Inductive step (for $n > 2$):

$$\begin{aligned}
&((n-1)\cdots(2)(1)+1) \bmod m = 1 \\
\rightarrow\ &(n(n-1)\cdots(2)(1)+n) \bmod m = n \bmod m \\
\rightarrow\ &(n(n-1)\cdots(2)(1)+1+(n-1)) \bmod m = n \bmod m \\
\rightarrow\ &(n(n-1)\cdots(2)(1)+1) \bmod m = (n-(n-1)) \bmod m \\
\rightarrow\ &(n(n-1)\cdots(2)(1)+1) \bmod m = 1
\end{aligned}$$

Inductive conclusion: By the principle of mathematical induction, $p(n)$ is true for all integers $n > 1$.

Supplementary Explanation

Answer. $(n!+1) \bmod m = 1$ for all the specified n, m.

Derivation/proof. Let $P(n)$ assert the statement for all $2 \le m \le n$. Base $n = 2$: $3 \bmod 2 = 1$. Assume $P(n-1)$ for $n > 2$. If $2 \le m \le n-1$, the hypothesis implies $(n-1)! \equiv 0 \pmod m$, hence $n! \equiv 0 \pmod m$. For the newly allowed $m = n$, $n! = n(n-1)!$ is directly divisible by n. In both cases $n! + 1 \equiv 1 \pmod m$, and $m \ge 2$ makes 1 the actual remainder. This fills the new-modulus case left implicit in the printed proof.

I021

Induction

2014 Final | Question 7 | Source page 11 | Difficulty 3 | 8 marks
Q-2014-FINAL-P07 | Answer provenance: official

> Let n be a positive integer. Use induction to prove that, for any non-negative integer m, there exist integers q and r such that
>
> $$m = qn + r, \qquad 0 \leq r < n.$$

Official Answer

Source: PDF physical page 11 (2014-final.pdf).

- **Proof by weak induction:**

Base Case: If $m < n$, we set $q = 0$ and $r = m$. Then (1) is true.

Induction Hypothesis: Assume (1) is true for the case of $m - 1$. That is, there exist q' and r' such that

$$m - 1 = q'n + r', \quad 0 \leq r' < n.$$

Inductive Step: Now consider the case of m. There are two cases:

(a) $r' < n - 1$. In the case, we set $q = q'$ and $r = r' + 1$. Then (1) is true.

(b) $r' = n - 1$. In the case, we set $q = q' + 1$ and $r = 0$. Then (1) is true.

By the principle of Mathematical Induction, we conclude that (1) is true for all non-negative integers m.

- **Proof by strong induction:**

Base Case: If $m < n$, we set $q = 0$ and $r = m$. Then (1) is true.

Induction Hypothesis: Assume (1) is true for any non-negative integer w such that $n \leq w < m$.

Inductive Step: Now consider the case of $n \leq m$. Let $w = m - n$. By the induction hypothesis, there exist q' and r' such that

$$w = q'n + r', \quad 0 \leq r' < n.$$

Hence,

$$m = w + n = (q' + 1)n + r'.$$

So, if we set $q = q' + 1$ and $r = r'$, then (1) is true.

By the principle of Mathematical Induction, we conclude that (1) is true for all non-negative integers m.

Supplementary Explanation

Answer. Such integers q, r exist for every $m \geq 0$.

Derivation/proof. For $m < n$, take $q = 0, r = m$. In a strong-induction step for $m \geq n$, let $w = m - n < m$. By the hypothesis $w = q'n + r'$ with $0 \leq r' < n$; then $m = (q' + 1)n + r'$, giving the required pair.

The complete strong-induction hypothesis covers all $0 \leq w < m$. Since $n > 0$, $0 \leq m - n < m$. The base interval therefore covers the possible small values of $m - n$.

I022

Induction

2013 Final | Question 7 | Source page 10 | Difficulty 3 | 10 marks
Q-2013-FINAL-P07 | Answer provenance: official

> Use induction to prove that, for any integer $n \geq 1$, $5^n + 2 \cdot 11^n$ is divisible by 3.

Official Answer

Source: PDF physical page 10 (2013-final.pdf).

Proof: Base case: When $n = 1$, we have

$$5^1 + 2 \times 11^1 = 27,$$

which is divisible by 3. So, the statement is true for $n = 1$.

Induction hypothesis: Now let $n > 1$. Assume the statement is true for $n - 1$, i.e.,

$$5^{n-1} + 2 \cdot 11^{n-1} \text{ is divisible by 3.}$$

Induction step: Consider the case of n:

$$\begin{aligned} 5^n + 2 \cdot 11^n &= 5^{n-1} + 4 \cdot 5^{n-1} + 2 \cdot 11^{n-1} + 20 \cdot 11^{n-1} \\ &= 5^{n-1} + 2 \cdot 11^{n-1} + 4(5^{n-1} + 5 \cdot 11^{n-1}) \\ &= 5^{n-1} + 2 \cdot 11^{n-1} + 4(5^{n-1} + 2 \cdot 11^{n-1} + 3 \cdot 11^{n-1}) \\ &= 3y + 4(3y + 3 \cdot 11^{n-1}) \\ &= 3(5y + 4 \cdot 11^{n-1}), \end{aligned}$$

which is divisible by 3.

By the principle of Mathematical Induction, we conclude that the statement is true for all integer $n \geq 1$. Q.E.D

Supplementary Explanation

Answer. The expression is divisible by 3 for all $n \geq 1$.

Derivation/proof. At $n = 1$, $5 + 2 \cdot 11 = 27$. Assume $5^{n-1} + 2 \cdot 11^{n-1} = 3u$. Then

$$5^n + 2 \cdot 11^n = (5^{n-1} + 2 \cdot 11^{n-1}) + 4(5^{n-1} + 5 \cdot 11^{n-1}) = 3u + 4(3u + 3 \cdot 11^{n-1}) = 3(5u + 4 \cdot 11^{n-1}).$$

The induction hypothesis therefore makes the next expression a multiple of three, completing the proof.

I023

Induction

2013 Final | Question 9 | Source page 12 | Difficulty 3 | 11 marks
Q-2013-FINAL-P09 | Answer provenance: official

> Consider a function $T(n)$ defined on integers n that are powers of 3. Suppose
>
> $$T(1) = 1, \qquad T(n) \leq 9T\left(\frac{n}{3}\right) + 4n^2 + 100n \quad \forall n > 1.$$
>
> Use advanced induction to prove that $T(n) = O(n^2 \log n)$.

Official Answer

Source: PDF physical page 12 (2013-final.pdf); PDF physical page 13 (2013-final.pdf).

Proof 1: It suffices to show that there exist n_0 and c such that

$$T(n) \leq cn^2 \log n \qquad \forall n = 3^i, n > n_0.$$

- Base case: Pick $n_0 = 1$. Then the base case is when $n = 3^1 = 3$:

  $$T(3) \leq 9T(1) + 4\,3^2 + 100 \times 3 = 345.$$

  We want the right hand side to be no greater than $c3^2 \log 3$. To satisfy the condition, we need $c \geq \frac{115}{3 \log 3}$.
- Induction Hypothesis: Suppose

  $$T(m) \leq cm^2 \log m \qquad \forall m = 3^i, m < n.$$

- Induction Step:

  $$\begin{aligned} T(n) &\leq 9T(\frac{n}{3}) + 4n^2 + 100n \\ &\leq 9c(\frac{n}{3})^2 \log \frac{n}{3} + 4n^2 + 100n \\ &= cn^2 \log n - cn^2 \log 3 + 4n^2 + 100n \\ &= cn^2 \log n \quad \text{if } c > 104/\log 3 \end{aligned}$$

  So, the proof follows through if we choose $n_0 = 1$ and $c = \max\{\frac{115}{3 \log 3}, 104/\log 3\} = 104/\log 3$. The proof is completed.

Proof 2: It suffices to show there exist positive constants c_1, c_2 and n_0 such that

$$T(n) \leq c_1 n^2 \log_3 n - c_2 n, \ \forall n > n_0. \qquad (*)$$

Base case: Let $n_0 = 1$. Remember n is a power of 3. The smallest such integer such that $n > 1$ is 3. So, the base case is $n = 3$. For this case, we have

$$T(3) \leq 9T(1) + 4 \times 3^2 + 100 \times 3 = 345 \leq 9c_1 - 3c_2.$$

This is true when $c_1 \geq \frac{3c_2 + 345}{9} = \frac{c_2 + 115}{3}$.

Induction hypothesis: Now let $n > 3$. Assume that, for any m such that $1 \leq m < n$ and m is power of 3, the following is true:

$$T(m) \leq c_1 m^2 \log_3 m - c_2 m.$$

Induction step: Now consider the case of n:

$$\begin{aligned}
T(n) &\leq 9T(\frac{n}{3}) + 4n^2 + 100n \\
&\leq 9(c_1(\frac{n}{3})^2 \log_3 \frac{n}{3} - c_2\frac{n}{3}) + 4n^2 + 100n \\
&= c_1 n^2 \log_3 n - c_1 n^2 - 3c_2 n + 4n^2 + 100n \\
&= c_1 n^2 \log_3 n - c_2 n + (4 - c_1)n^2 + (100 - 2c_2)n \\
&\leq c_1 n^2 \log_3 n - c_2 n + (4 - c_1)n^2 + (100 - 2c_2)n^2 \\
&= c_1 n^2 \log_3 n - c_2 n + (104 - c_1 - 8c_2)n^2 \\
&\leq c_1 n^2 \log_3 n - c_2 n, \quad \text{when } (104 - c_1 - 2c_2)n^2 \leq 0
\end{aligned}$$

So, we have shown that inequality (*) is true when

$$n_0 = 1, c_2 > 0, c_1 \geq \max\{\frac{c_2 + 115}{3}, 104 - 2c_2\}.$$

Consequently, $T(n) = O(n^2 \log n)$. Q.E.D

Supplementary Explanation

Answer. $T(n) \leq cn^2 \log n$ for suitable c; thus $T(n) = O(n^2 \log n)$.

Derivation/proof. For $n = 3$, $T(3) \leq 345$, so take $c \geq 115/(3 \log 3)$. Assume $T(m) \leq cm^2 \log m$ for smaller powers. Then $T(n) \leq 9c(n/3)^2 \log(n/3) + 4n^2 + 100n = cn^2 \log n - cn^2 \log 3 + 4n^2 + 100n \leq cn^2 \log n$ when $c \geq 104/\log 3$ (using $100n \leq 100n^2/3$ for $n \geq 3$).

For powers of three with $n \geq 9$, the induction hypothesis applies to $n/3 \geq 3$. Taking $c = 104/\log 3$ gives $-cn^2 \log 3 = -104n^2$, while $4n^2 + 100n \leq 104n^2$. The estimate need not hold at $n = 1$, where $\log 1 = 0$.

The base calculation is $T(3) \leq 9 + 36 + 300 = 345$.

I024

Induction

2025 Spring Midterm | Question 3 | Source page 5 | Difficulty 4 | 12 marks
Q-2025S-MID-P03 | Answer provenance: official

> Take the natural numbers $\mathbb{N}$ as the domain. Recall that in Peano's axiomatic system, addition is defined as follows: (i) $x + 0 = x$; (ii) $x + S(y) = S(x + y)$. We can further define the predicate $x \leq y$ as $\exists k(x + k = y)$. Prove that for any $x, y \in \mathbb{N}$, either $x \leq y$ or $y \leq x$ is true. You are only allowed to use Peano's axioms (including induction), definitions, and the commutative law of addition.

Official Answer

Source: physical PDF page 5 (COMP2711_S2025_Midterm.pdf).

Proof. We will use induction on x to prove the statement.

Base case $x = 0$: we will show that for any $y \in \mathbb{N}$, $0 \leq y$. By the definition of $\leq$, $0 \leq y$ means $\exists k(0 + k = y)$. Taking $k = y$, we have $0 + y = y + 0 = y$, so this is true.

Inductive step: Assume that for some $x \in \mathbb{N}$, the statement holds, i.e., for any $y \in \mathbb{N}$, either $x \leq y$ or $y \leq x$. We will prove that the statement also holds for $S(x)$, i.e., for any $y \in \mathbb{N}$, either $S(x) \leq y$ or $y \leq S(x)$.

Case 1: $y = 0$: In this case, we can show that $y \leq S(x)$ following the same argument as above.

Case 2: $y \neq 0$: Since $y \neq 0$, $y = S(y')$ for some $y' \in \mathbb{N}$ (by Peano's axioms). By the inductive hypothesis, either $x \leq y'$ or $y' \leq x$.

- If $x \leq y'$, then $\exists k(x + k = y')$. Applying S to both sides gives $S(x + k) = S(y')$. By the commutative law and the definition of addition, $S(x + k) = S(k + x) = k + S(x) = S(x) + k$. This means $S(x) \leq S(y')$, i.e., $S(x) \leq y$.
- If $y' \leq x$, then $\exists k(y' + k = x)$. Applying S to both sides gives $S(y' + k) = S(x)$. By the commutative law and the definition of addition, $S(y' + k) = S(k + y') = k + S(y') = S(y') + k$. This means $S(y') \leq S(x)$, i.e., $y \leq S(x)$.

Supplementary Explanation

Answer. $\forall x \forall y(x \leq y \vee y \leq x)$.

Derivation/proof. First, induction shows that every natural number is zero or a successor: zero satisfies the disjunction, and any $S(z)$ is itself a successor. Now induct on x. For $x = 0$, arbitrary y satisfies $0 + y = y + 0 = y$, so $k = y$ witnesses $0 \leq y$.

Assume that for every y we have $x \leq y$ or $y \leq x$. Compare $S(x)$ with arbitrary y. If $y = 0$, choose $k = S(x)$ to obtain $0 \leq S(x)$. Otherwise write $y = S(y')$ and apply the hypothesis to y'. If $x + k = y'$, then $S(x) + k = k + S(x) = S(k + x) = S(x + k) = S(y') = y$, so $S(x) \leq y$. If $y' + k = x$, the same identities give $S(y') + k = S(y' + k) = S(x)$, so $y \leq S(x)$. All cases are covered, establishing the claim by induction without assuming a preexisting total order.

I025

Induction

2023 Fall Final | Question 8(a) | Source page 12 | Difficulty 4 | 7 marks
Q-2023F-FINAL-P08-a | Answer provenance: official

> For every non-negative integer n, a_{n+1} is calculated from a_n with the following rules:
>
> If the rightmost digit of a_n is bigger than 5, then $a_{n+1} = 9 \times a_n$. Otherwise, we just "remove" the rightmost digit of a_n and the remaining number is a_{n+1}. If a_{n+1} does not contain any digits we stop.
>
> As an example, if $a_n = 124$ then $a_{n+1} = 12$ as a_n's rightmost digit is 4 and we simply remove it from 124 to form 12. Also if $a_n = 16$ then $a_{n+1} = 9 \times 16 = 144$ as a_n's rightmost digit was bigger than 5.
>
> Use strong induction on a_0 to prove that this process stops in finitely many steps for any arbitrary a_0. Please write a complete induction proof consisting of base case or cases, inductive hypothesis, inductive step and conclusion.
>
> (Hint 1: If this process stops for any $1 \leq a_0 \leq k$, can you show it will stop in finitely many steps for $a_0 = k + 1$ as well?)
>
> (Hint 2: If the rightmost digit of a number a_n is larger than 5, what can we say about the possible values for the rightmost digit of $9 \times a_n$?)

Official Answer

Source: physical PDF page 12 (COMP2711_F2023_Final.pdf).

> (a) We use strong induction on the first element a_0. Obviously this process stops in 1 step if $a_0 = 1$. Assume that this assumption is correct for all values between 1 up to k. And now we prove it will be also correct for $k + 1$. If $K + 1$ ends with a digit that is smaller or equal than 5 then we just simply remove that digit and the result number is already smaller than $K + 1$ so it will stop after finitely many steps. If $K + 1$ ends with a digit that is bigger than 5, then We get $9(k + 1)$ as the next number that definitely ends with a digit smaller than 6. So for its next step we need to remove the rightmost digit that will give us a number that is again smaller than $k + 1$. (because $9(k + 1)/10$ is smaller than $k + 1$ for sure)

Supplementary Explanation

Answer. The process terminates for every nonnegative integer starting value.

Derivation/proof. For the single-digit starts 0 through 5, deletion immediately removes the only digit and terminates. Use strong induction on a positive starting integer N; assume every smaller nonnegative starting value terminates. If N ends in a digit at most 5, deletion either stops immediately or leaves $\lfloor N/10 \rfloor < N$, which terminates by the hypothesis. If its last digit is $r \in \{6, 7, 8, 9\}$, the last digit of $9N$ is $10 - r \in \{4, 3, 2, 1\}$. The next step therefore deletes that digit and leaves $\lfloor 9N/10 \rfloor < N$. This smaller value terminates by the induction hypothesis. In both cases only one or two steps precede a terminating smaller instance, proving the result for N and hence for every allowed start.

Base cases $N = 0, 1, 2, 3, 4, 5$ terminate after deleting the only digit. For $N \geq 6$, use the induction hypothesis for all smaller nonnegative starting values. If the last digit exceeds five, the two-step value is $\lfloor 9N/10 \rfloor < N$. Thus the induction step is complete.

I026

Induction

2021 Final Part 1 | Question 3 | Source page 2 | Difficulty 4 | 10 marks
Q-2021-FINAL1-P03 | Answer provenance: official

> Consider a company with $n > 0$ workers. A worker is a famous-solo-worker if she or he is known by every other worker, but knows none of them. You are the boss and you want to check if there is a famous-solo-worker in your company. You are only allowed to ask the following question to a worker: "Do you know worker j?" The workers will answer your questions truthfully. Show that you can find the famous-solo-worker, or declare that such a worker doesn't exist, with at most $3(n-1)$ questions.
>
> Note that there can be at most one famous-solo-worker.

Official Answer

Source: physical PDF page 2 (2021-final-part-1.pdf), physical PDF page 3 (2021-final-part-1.pdf).

> We prove this by induction.
>
> **Basis step:** When $n = 1$, the only worker must be a famous-solo-worker, and no questions are needed, which equals to $3(1-1) = 0$.
>
> **Inductive step:** The inductive hypothesis is that if there are $k \geq 1$ workers, then we can determine whether there is a famous-solo-worker with at most $3(k-1)$ questions.
>
> Suppose there are $k+1$ workers. Let i and j be any two workers among them. We ask whether i knows j, this takes one question. If i knows j, then i is not a famous-solo-worker. If i doesn't know j, then j is not a famous-solo-worker. Without loss of generality, assume that i is not a famous-solo-worker. Excluding i from the $k+1$ workers, there are k workers (including j) remaining. By the inductive hypothesis, we can use $3(k-1)$ questions for our checking.
>
> If there is no famous-solo-worker among these k workers, then we know that there is no famous-solo-worker, and the number of questions used is $1 + 3(k-1) = 3k - 2 \leq 3k$.
>
> If there is a famous-solo-worker s among these k workers, then we ask whether s knows i and whether i knows s. This takes two more questions. If i knows s and s doesn't know i, then s is still a famous-solo-worker, otherwise s is not a famous-solo-worker (thus no famous-solo-worker). In this case, we used a total of $1 + 3(k-1) + 2 = 3k$ questions.

Supplementary Explanation

Answer. Eliminate candidates with $n-1$ questions, then verify the survivor using at most $2(n-1)$ more.

Derivation/proof. Start with any candidate c. For each other worker j ask whether c knows j. If yes, c is disqualified and replace c by j; if no, j is disqualified because c does not know j. An induction over comparisons shows any genuine famous-solo-worker remains among the candidates, so n−1 questions leave only c possible. Ask c whether they know each other worker and ask each other worker whether they know c, using at most 2(n−1) questions. Accept c iff all first answers are no and all second answers yes; otherwise nobody qualifies, since all others were already disqualified. For n=1 both conditions are vacuous and zero questions suffice.

The iterative proof below uses at most $(n-1) + 2(n-1) = 3(n-1)$ questions. Each eliminated person has a concrete disqualifying answer, so failure of final verification establishes nonexistence rather than merely failure of the candidate.

Elimination never eliminates a genuine famous-solo-worker. Accept the final candidate exactly when all first-direction answers are no and all reverse-direction answers are yes.

I027

Induction

2020 Final Part 1 | Question 2 | Source page 1 | Difficulty 4 | 8 marks
Q-2020-FINAL1-P02 | Answer provenance: official

> Prove that for any $n \in \mathbb{N}$, there exists an n-digit number A_n such that all its digits are either 1 or 2, and $2^n \mid A_n$. [Hint: Use induction.]

Official Answer

Source: physical PDF page 1 (2020-final-part-1.pdf); physical PDF page 2 (2020-final-part-1.pdf).

We use induction on n. Base case: $n = 1$. We can set $A_1 = 2$, $2^1 \mid A_1$. Induction hypothesis: Assume it is correct for $n = k$, which means that there exists A_k such that $2^k \mid A_k$. Now consider $n = k + 1$: As $2^k \mid A_k$, there are two possibilities for the remainder of A_k dividing 2^{k+1}. $A_k = 2^k \cdot l$, where l is even, which means $A_k \equiv 0 \pmod{2^{k+1}}$, or $A_k = 2^k \cdot (1 + l')$, where l' is even, which means $A_k \equiv 2^k \pmod{2^{k+1}}$.

1. $A_k \equiv 0 \pmod{2^{k+1}}$. In this scenario we add the digit 2 to the left side of A_k. So $A_{k+1} = \overline{2A_k}$. We show that $2^{k+1} \mid A_{k+1}$. $A_{k+1} = 2 \cdot 10^k + A_k = 2 \cdot 2^k \cdot 5^k + A_k = 2^{k+1} \cdot 5^k + A_k$. And we know that $2^{k+1} \cdot 5^k \equiv 0 \pmod{2^{k+1}}$ and $A_k \equiv 0 \pmod{2^{k+1}}$. So $A_{k+1} \equiv 0 \pmod{2^{k+1}}$.
2. $A_k \equiv 2^k \pmod{2^{k+1}}$. In this scenario we add the digit 1 to the left side of A_k. So $A_{k+1} = \overline{1A_k}$. We show that $2^{k+1} \mid A_{k+1}$. $A_{k+1} = 1 \cdot 10^k + A_k = 2^k \cdot 5^k + A_k$. And we know that $A_k \equiv 2^k \pmod{2^{k+1}}$. So $A_{k+1} \equiv 2^k + 2^k \cdot 5^k \equiv 2^k \cdot (1 + 5^k) \pmod{2^{k+1}}$. $1 + 5^k$ is an even number so $A_{k+1} \equiv 0 \pmod{2^{k+1}}$.

Supplementary Explanation

Answer. Such a number exists for every positive digit length n.

Derivation/proof. Base n=1: take A1=2. Suppose Ak has k digits in {1,2} and $A_k = 2^k t$. Choose a leading digit d∈{1,2} with d+t even: d=2 when t is even and d=1 when t is odd. Set $A_{k+1} = d10^k + A_k = 2^k(d5^k + t)$. Because 5^k is odd, the parenthesis is even by the choice of d. Hence $2^{k+1} \mid A_{k+1}$. Adding a nonzero leading digit creates exactly k+1 digits without changing the allowed suffix digits, completing induction.

The source notation $\overline{2A_k}$ and $\overline{1A_k}$ denotes digit concatenation, not multiplication.

The initial value is $A_1 = 2$. If $A_k = 2^k t$ with integer t, the recurrence supplies one more factor of two in the next term; this establishes the divisibility induction.

I028

Induction

2019 Final | Question Bonus Problem(a) | Source page 17 | Difficulty 4 | 3 marks
Q-2019-FINAL-BONUS-a | Answer provenance: official

> Consider the following recurrence: $f_1 = 1$, $f_n = 2f_{n-1} + n$ for $n \geq 2$. The following is a purported induction proof that $f_n = O(n)$.
>
> Proof. Base case: When $n = 1$, $f(n) = O(n)$ because $f(1) = O(1)$.
>
> Induction step: Assuming $f_{n-1} = O(n-1)$, we will prove $f_n = O(n)$:
>
> $$\begin{aligned} f_n &= 2f_{n-1} + n \\ &= 2 \cdot O(n-1) + n && \text{(induction hypothesis)} \\ &= O(3n-2) \\ &= O(n) && \text{(dropping lower order term and constant coefficient)} \end{aligned}$$
>
> as desired.
>
> (a) Identify the mistake in the proof.

Official Answer

Source: physical PDF page 17 (2019-final.pdf).

The hidden constant in the induction conclusion $f_n = O(n)$ is different (larger) from the one in the induction hypothesis $f_{n-1} = O(n-1)$. Generally, one cannot use asymptotic notation in induction proofs, because the induction step is implicitly applied many times, so the constant coefficient will no longer be a constant.

Supplementary Explanation

Answer. The induction silently changes the hidden big-O constant instead of preserving one constant for all n.

Derivation/proof. An actual claim $f_n \leq Cn$ needs one fixed C and a threshold. From the induction hypothesis $f_{n-1} \leq C(n-1)$, the recurrence gives only $f_n \leq 2C(n-1) + n = (2C+1)n - 2C$. This is generally not bounded by the same Cn, since their difference is $(C+1)n - 2C$, positive for large n. Writing O again hides this growth of constants. A statement about the single value f(1) also cannot by itself establish an asymptotic bound.

A value at the single input $n = 1$ is also not by itself a valid base case for an asymptotic statement.

The same constant must work throughout the proposed induction. The unstrengthened estimate would require $(C+1)n \leq 2C$ for arbitrarily large n, which fails for each fixed C; this explains why that particular induction bound cannot close.

I029

Induction

2019 Final | Question Bonus Problem(b) | Source page 17 | Difficulty 4 | 7 marks
Q-2019-FINAL-BONUS-b | Answer provenance: official

> Consider the following recurrence: $f_1 = 1$, $f_n = 2f_{n-1} + n$ for $n \geq 2$. The following is a purported induction proof that $f_n = O(n)$.
>
> Proof. Base case: When $n = 1$, $f(n) = O(n)$ because $f(1) = O(1)$.
>
> Induction step: Assuming $f_{n-1} = O(n-1)$, we will prove $f_n = O(n)$:
>
> $$\begin{aligned} f_n &= 2f_{n-1} + n \\ &= 2 \cdot O(n-1) + n && \text{(induction hypothesis)} \\ &= O(3n-2) \\ &= O(n) && \text{(dropping lower order term and constant coefficient)} \end{aligned}$$
>
> as desired.
>
> (b) Prove that $f(n) = O(2^n)$ by induction.

Official Answer

Source: physical PDF page 17 (2019-final.pdf).

We will show that $f(n) \leq 2 \cdot 2^n - n - 2$. Base case: When $n = 1$, $f(n) = 1 \leq 2 \cdot 2^1 - 1 - 2$. Induction step: Assuming $f_{n-1} \leq 2 \cdot 2^{n-1} - (n-1) - 2$, we will prove $f_n \leq 2 \cdot 2^n - n - 2$:

$$\begin{aligned} f_n &= 2f_{n-1} + n \\ &\leq 2(2 \cdot 2^{n-1} - (n-1) - 2) + n && \text{(induction hypothesis)} \\ &= 2 \cdot 2^n - n - 2. \end{aligned}$$

Comment: It is impossible to prove $f_n \leq c \cdot 2^n$ by induction, no matter how you choose the constant c. Interestingly, and somehow counter-intuitively, it is possible to prove the stronger claim $f(n) \leq c \cdot 2^n - n - 2$. Fundamentally, this is because a stronger claim will also make the induction hypothesis stronger, which sometimes makes the induction step easier to go through. This technique is known as "strengthening the induction hypothesis".

Supplementary Explanation

Answer. $f_n = 2^{n+1} - n - 2 \leq 2 \cdot 2^n$, so $f_n = O(2^n)$.

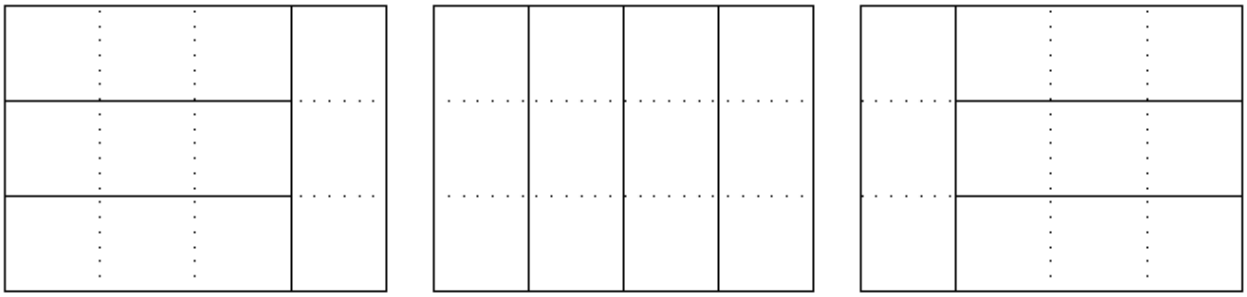
Derivation/proof. Prove the stronger exact identity by induction. At n=1, $2^2 - 1 - 2 = 1 = f_1$. Assume $f_{n-1} = 2^n - (n-1) - 2$. Then $f_n = 2[2^n - (n-1) - 2] + n = 2^{n+1} - n - 2$. Thus the identity holds for all $n \geq 1$, and immediately gives the fixed bound $f_n \leq 2 \cdot 2^n$. Equivalently, the source's strengthened inequality $f_n \leq 2 \cdot 2^n - n - 2$ propagates by the same algebra. Directly iterating the recurrence verifies its first values $1, 4, 11, 26$, matching the formula.

The first four values $1, 4, 11, 26$ satisfy both the recurrence and formula. Positivity together with $f_n \leq 2 \cdot 2^n$ gives fixed constant $C = 2$ and threshold $n_0 = 1$.

I030

Recursion

2018 Final | Question 2(a) | Source page 4 | Difficulty 1
Q-2018-FINAL-P02-a-E1 | Answer provenance: official

> Let E_n be the number of ways in which you can tile a 3-by-n board with 1-by-3 pieces. For example, when $n = 4$ the board is 3-by-4 and $E_4 = 3$. The 3 ways are shown as follows:
>
> 
>
> (a) Calculate E_1, E_2, and E_3. [Atomic request: E_1.]

Official Answer

Source: physical PDF page 4 (2018-final.pdf).

1, 1, 2.

Supplementary Explanation

Answer. $E_1 = 1$.

Derivation/proof. The board has three cells in one column. A horizontal piece of length three cannot fit in width one, so the only possible placement is a single vertical piece covering all three cells. This placement completes the board and is unique.

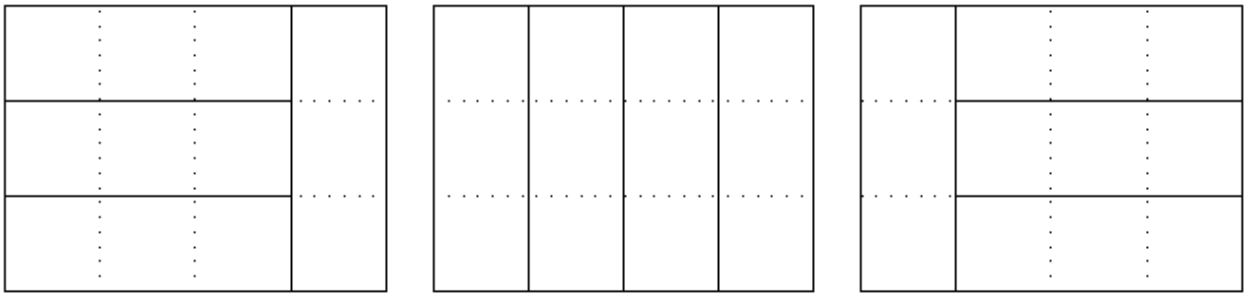
The three printed values are respectively E_1, E_2, E_3; this atomic question asks for the first.

A 1×3 tile covers the entire column in the one-column board. This is the sole tiling and gives the first initial value.

I031

Recursion

2018 Final | Question 2(a) | Source page 4 | Difficulty 1
Q-2018-FINAL-P02-a-E2 | Answer provenance: official

> Let E_n be the number of ways in which you can tile a 3-by-n board with 1-by-3 pieces. For example, when $n = 4$ the board is 3-by-4 and $E_4 = 3$. The 3 ways are shown as follows:
>
> 
>
> (a) Calculate E_1, E_2, and E_3. [Atomic request: E_2.]

Official Answer

Source: physical PDF page 4 (2018-final.pdf).

1, 1, 2.

Supplementary Explanation

Answer. $E_2 = 1$.

Derivation/proof. A horizontal piece requires three columns but this board has only two. Thus both columns must be filled vertically, one piece per column. Those two placements cover every cell and there is no choice about their positions, so only one tiling exists.

The three printed values are respectively E_1, E_2, E_3; this atomic question asks for the second.

In a two-column board, each of the two columns must be covered vertically: a horizontal tile would need three columns. These two tiles cover its six cells uniquely.

I032

Recursion

2018 Final | Question 4(c) | Source page 6 | Difficulty 1
Q-2018-FINAL-P04-c-basis | Answer provenance: official

> For this question, no proof is required. Fill in the following recursive definition so that S is the set of complete graphs.
>
> Recursive step: Suppose that $G = (V, E)$ is in S, where $V = \{v_1, v_2, \ldots, v_n\}$. Then $G' = (V', E')$ is in S, where $V' = \{v_1, v_2, \ldots, v_n, v_{n+1}\}$.
>
> (c) Fill in the following recursive definition so that S is the set of complete graphs.
>
> Basis: ____ is in S.
>
> [Atomic request: basis graph; no proof is required.]

Official Answer

Source: physical PDF page 6 (2018-final.pdf).

Basis: The graph with one vertex and no edges is a complete graph.

Supplementary Explanation

Answer. K_1, the graph with one vertex and no edges.

Derivation/proof. A single vertex has no pair of distinct vertices, so the condition that every such pair is adjacent holds vacuously. It is therefore a complete graph. Adding one vertex at a time from this basis can reach every positive number of vertices, whereas starting with K_2 would omit K_1.

Start with a single vertex with no edges. Adding one new vertex adjacent to every old vertex successively constructs $K_2, K_3, \ldots$; no self-loop is added.

I033

Recursion

2017 Final | Question 10(a) | Source page 14 | Difficulty 1
Q-2017-FINAL-P10-a-first | Answer provenance: official

> Let S be the subset of the set of ordered pairs of integers defined recursively by:
>
> Basis step: $(0, 0) \in S$
>
> Recursive step: If $(a, b) \in S$, then $(a, b + 1) \in S$, $(a + 1, b + 1) \in S$, and $(a + 2, b + 1) \in S$.
>
> (a) List the elements of S produced by the first two applications of the recursive definition. [Atomic request: first application.]

Official Answer

Source: physical PDF page 14 (2017-final.pdf).

First Application of Recursive step to S Applying recursive step to $(0, 0)$ gives $(0, 1), (1, 1), (2, 1)$. So $S = \{(0, 0), (0, 1), (1, 1), (2, 1)\}$.

Supplementary Explanation

Answer. New elements: $(0, 1), (1, 1), (2, 1)$; the accumulated set also contains $(0, 0)$.

Derivation/proof. Substitute $a = b = 0$ into the three rules. They give $(0, 0 + 1)$, $(0 + 1, 0 + 1)$, and $(0 + 2, 0 + 1)$, namely the three displayed pairs. None equals the basis pair because the second coordinate has increased to one.

Collecting the listed ordered pairs as a set gives exactly $\{(0, 0), (0, 1), (1, 1), (2, 1)\}$; the second coordinate alone need not determine the first.

I034

Recursion

2017 Final | Question 11(a) | Source page 15 | Difficulty 1
Q-2017-FINAL-P11-a-T1 | Answer provenance: official

> In a Double Towers of Hanoi problem, there are three poles in a row and $2n$ disks for some positive integer n. There are two disks of size 1, two disks of size 2, and so on, up to two disks of size n. Initially one of the poles contains all the disks placed on top of each other in pairs of decreasing size. Disks are transferred one-by-one from one pole to another, but at no time may a larger disk be placed on top of a smaller disk. A disk may be placed on top of one of the same size. Let $T(n)$ be the number of moves needed to transfer a tower of $2n$ disks from one pole to another.
>
> (a) Find $T(1)$ and $T(2)$. [Atomic request: $T(1)$.]

Official Answer

Source: physical PDF page 15 (2017-final.pdf).

$T(1) = 2$.

Supplementary Explanation

Answer. $T(1) = 2$.

Derivation/proof. There are only two disks, of the same size. Move the top disk from the source to the destination, then place the second disk on top of it. Both moves are legal because equal disks may stack. Each disk must move at least once, so two is also a lower bound.

Both same-size disks must change poles and one move changes only one disk. Hence at least two moves are necessary, and moving them consecutively achieves that lower bound.

I035

Recursion

2025 Fall Final | Question 10(i) | Source page 17 | Difficulty 2 | 2 marks
Q-2025F-FINAL-P10-i | Answer provenance: official

> A spider starts with 10 legs. Each day, its legs instantly double by splitting, but immediately after, exactly 2 legs fall off and are lost forever. Let a_n be the number of legs after n days, with $a_0 = 10$.
>
> (i) [2 points] Write the recurrence relation for a_n.

Official Answer

Source: physical PDF page 17 (COMP2711_F2025_Final.pdf).

(i) The recurrence relation is

$$a_n = 2a_{n-1} - 2 \quad (n \geq 1), \qquad a_0 = 10.$$

Supplementary Explanation

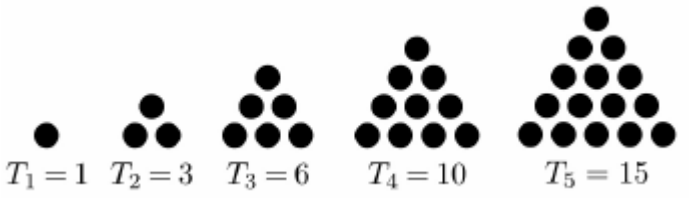
Answer. $a_0 = 10$ and $a_n = 2a_{n-1} - 2$ for $n \geq 1$.

Derivation/proof. At the end of day $n-1$ there are a_{n-1} legs. On day n, each splits into two, giving $2a_{n-1}$ legs; exactly two then fall off. Hence $a_n = 2a_{n-1} - 2$. This update starts on day one, so $n \geq 1$, while the given initial value is $a_0 = 10$. On the first day, $a_1 = 20 - 2 = 18$, agreeing with the stated order of events.

I036

Recursion

2024 Fall Final | Question 9(b)(i) | Source page 24 | Difficulty 2 | 1 mark
Q-2024F-FINAL-P09-b-i | Answer provenance: official

> Consider the following picture:
>
> 
>
>
> Consider the sequence of triangles formed by circles. Let $T(n)$ be the number of circles needed to form a triangle with n circles on each side.
>
> (i) Provide a recurrence relation for $T(n)$.

Official Answer

Source: physical PDF page 24 (COMP2711_F2024_Final.pdf).

> (b) (i) $T(1) = 1, \quad T(n) = T(n-1) + n$

Supplementary Explanation

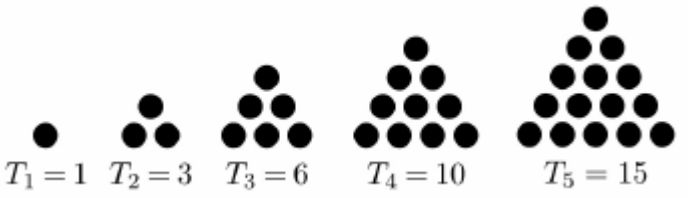
Answer. $T(1) = 1, \quad T(n) = T(n-1) + n \ (n \geq 2)$.

Derivation/proof. The side-one triangle has a single circle, so $T(1) = 1$. For side length $n \geq 2$, remove the bottom row containing exactly n circles. The remaining triangle has side length $n-1$ and $T(n-1)$ circles. The parts are disjoint and cover the full triangle, so $T(n) = T(n-1) + n$. Substituting gives $1, 3, 6, 10, 15$, matching the five pictured triangles.

I037

Recursion

2024 Fall Final | Question 9(b)(ii) | Source page 24 | Difficulty 2 | 2 marks
Q-2024F-FINAL-P09-b-ii | Answer provenance: official

> Consider the following picture:
>
> 
>
>
> Consider the sequence of triangles formed by circles. Let $T(n)$ be the number of circles needed to form a triangle with n circles on each side.
>
> (ii) Calculate the closed form formula for $T(n)$

Official Answer

Source: physical PDF page 24 (COMP2711_F2024_Final.pdf).

> (ii) $T(n) = n + T(n-1) = n + (n-1) + T(n-2) = \cdots = n + (n-1) + (n-2) + \cdots + 1 = \frac{n \cdot (n+1)}{2}$.

Supplementary Explanation

Answer. $T(n) = n(n+1)/2$.

Derivation/proof. Use $T(1) = 1, T(n) = T(n-1) + n$. Repeated substitution gives $T(n) = n + (n-1) + T(n-2) = \cdots = n + (n-1) + \cdots + 1$. Call this sum S and also write $S = 1 + 2 + \cdots + n$. Adding term by term gives n copies of $n+1$, so $2S = n(n+1)$ and $S = n(n+1)/2$. This is the desired closed form; at $n = 1$ it gives one, agreeing with the initial value.

I038

Recursion

2023 Fall Final | Question 8(b) | Source page 12 | Difficulty 2 | 4 marks
Q-2023F-FINAL-P08-b | Answer provenance: official

> Calculate the closed form of the following recursive relation:
>
> $$a_0 = 1, \qquad a_n = 3a_{n-1} + 2 \quad (\text{for } n \geq 1).$$

Official Answer

Source: physical PDF page 12 (COMP2711_F2023_Final.pdf).

> (b) $a_0 = 1 \quad b = 3 \quad c = 2$
>
> $$f_n = 1 \cdot 3^n + 2\frac{3^n - 1}{3 - 1}.$$

Supplementary Explanation

Answer. $a_n = 2 \cdot 3^n - 1$.

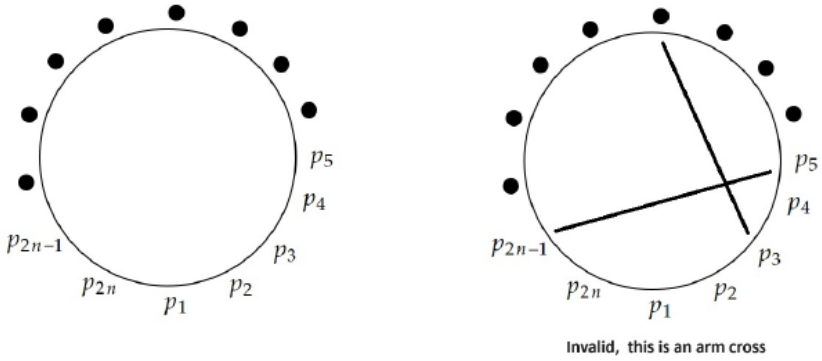
Derivation/proof. Define $b_n = a_n + 1$. Then $b_n = 3a_{n-1} + 3 = 3(a_{n-1} + 1) = 3b_{n-1}$ and $b_0 = 2$. Repeated substitution gives $b_n = 2 \cdot 3^n$, hence $a_n = 2 \cdot 3^n - 1$. Verify the initial condition: $2 \cdot 3^0 - 1 = 1$. For n≥1, $3(2 \cdot 3^{n-1} - 1) + 2 = 2 \cdot 3^n - 1$, confirming the formula satisfies the given recurrence for every index.

Let $b_n = a_n + 1$. Then $b_n = 3a_{n-1} + 3 = 3b_{n-1}$ with $b_0 = a_0 + 1 = 2$. Hence $a_n = 2 \cdot 3^n - 1$, and substitution gives $3(2 \cdot 3^{n-1} - 1) + 2 = 2 \cdot 3^n - 1$.

I039

Recursion

2023 Fall Final | Question 9(a) | Source page 14 | Difficulty 2 | 2 marks
Q-2023F-FINAL-P09-a | Answer provenance: official

> Assume that there are $2n$ people sitting around a round table. We want to split them into n pairs to engage in handshakes so that no arms cross, given that nobody is allowed to shake hands with more than one person at once. Let a_{2n} denote the number of ways that we can pair them legally without any arm crossing. You can see an invalid handshake in the figure below. Please answer the following questions by referring to the figure:
>
> 
>
>
> Is it possible to pair p_1, p_5 together for a handshake? Please explain why in at most 3 lines.

Official Answer

Source: physical PDF page 14 (COMP2711_F2023_Final.pdf).

> (a) No, because then on their right side they have $\{p_2, p_3, p_4\}$ which one of them has be paired with someone from the other side and this cause cross arm.

Supplementary Explanation

Answer. No.

Derivation/proof. The chord p_1p_5 leaves the three people p_2, p_3, p_4 on one side. An odd set cannot be perfectly paired internally, so one must pair with someone on the other side. That handshake crosses p_1p_5, contradicting legality.

This three-sentence proof uses the full-pairing requirement in the shared prompt; allowing an unmatched person would be a different problem.

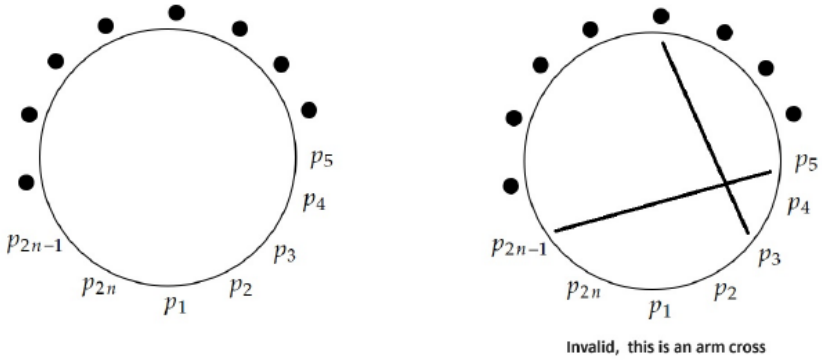
The side containing p_2, p_3, p_4 cannot be internally paired because three is odd. A pairing to the other side violates the noncrossing condition.

I040

Recursion

2023 Fall Final | Question 9(b) | Source page 14 | Difficulty 2 | 2 marks
Q-2023F-FINAL-P09-b | Answer provenance: official

Assume that there are $2n$ people sitting around a round table. We want to split them into n pairs to engage in handshakes so that no arms cross, given that nobody is allowed to shake hands with more than one person at once. Let a_{2n} denote the number of ways that we can pair them legally without any arm crossing. You can see an invalid handshake in the figure below. Please answer the following questions by referring to the figure:



Assume that you want to first pair p_1 with someone. List all potential candidates who can be paired with p_1 initially?

Official Answer

Source: physical PDF page 14 (COMP2711_F2023_Final.pdf).

> (b) It can only be paired with p_j where j is even. Otherwise the number of people in one side is odd and will guarantee an arm cross.

Supplementary Explanation

Answer. $p_2, p_4, p_6, \ldots, p_{2n}$.

Derivation/proof. If p_1 pairs with p_j, the intervening people $p_2, \ldots, p_{j-1}$ number $j-2$, and the other side has $2n-j$ people. A legal complete pairing cannot connect the two sides across the fixed chord, so both counts must be even; this is equivalent to j being even. Conversely, for each even j, pair consecutive people on each side. Those adjacent chords stay in their own sides and do not cross, so every listed candidate is feasible.

The two side sizes are $j-2$ and $2n-j$. If j is even, pair adjacent people on each side to construct a valid completion. This includes an empty side for $j=2$ or $j=2n$.

I041

Recursion

2020 Final Part 1 | Question 5(a) | Source page 3 | Difficulty 2
Q-2020-FINAL1-P05-a-formula | Answer provenance: official

> Consider the following series of functions defined recursively: $f_0(x) = x^{-1}$; for $n \geq 0$, $f_{n+1}(x) = f_n'(x)$ (recall that $(ax^b)' = abx^{b-1}$).
>
> Find the general formula of $f_n(x)$.

Official Answer

Source: physical PDF page 4 (2020-final-part-1.pdf).

$f_n(x) = (-1)^n n! \cdot x^{-(n+1)}$. We can prove this by induction. First $f_0(x) = (-1)^0 \cdot 0! \cdot x^{-1} = x^{-1}$. Suppose for some $k \geq 0$, $f_k(x) = (-1)^k k! \cdot x^{-(k+1)}$, then

$$f_{k+1}(x) = f\prime_k(x) = (-1)^k k! \cdot [-(k+1)] \cdot x^{-(k+2)} = (-1)^{k+1}(k+1)! \cdot x^{-(k+2)}.$$

Supplementary Explanation

Answer. $f_n(x) = (-1)^n n! x^{-(n+1)}$ for $x \neq 0$.

Derivation/proof. The first terms are x^{-1}, $-x^{-2}$, $2x^{-3}$, $-6x^{-4}$. At differentiation number j the current exponent is -j, so the coefficient is multiplied by -j and the exponent decreases by one. After n applications the coefficient is $(-1)(-2)\cdots(-n) = (-1)^n n!$ and exponent is -(n+1). The x≠0 domain is inherited from fo.

The first four functions are x^{-1}, $-x^{-2}$, $2x^{-3}$, $-6x^{-4}$. The condition $x \neq 0$ comes from the original function.

I042

Recursion

2019 Final | Question 6(a) | Source page 8 | Difficulty 2 | 2 marks
Q-2019-FINAL-P06-a | Answer provenance: official

> (a) Give a recursive definition of the function $ones(s)$, which counts the number of ones in a bit string s.
>
> (b) Use structural induction to prove that $ones(st) = ones(s) + ones(t)$.
>
> (a) Give a recursive definition of the function $ones(s)$, which counts the number of ones in a bit string s.

Official Answer

Source: physical PDF page 8 (2019-final.pdf).

Let λ denote an empty bit string. $\text{ones}(\lambda) = 0$ and $\text{ones}(wx) = x + \text{ones}(w)$, where w is a bit string and x is a bit (viewed as an integer when being added).

Supplementary Explanation

Answer. $\text{ones}(\lambda) = 0$; $\text{ones}(wx) = \text{ones}(w) + x$ for a bit $x \in \{0, 1\}$.

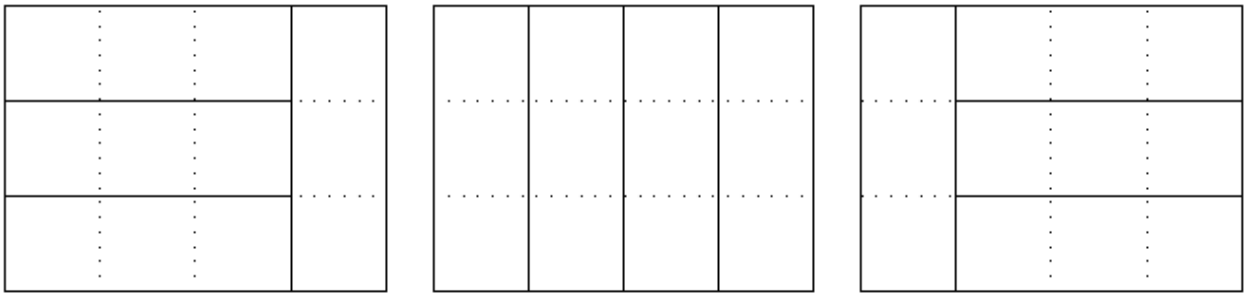
Derivation/proof. The empty string has no ones. Appending a zero leaves the count unchanged, and appending a one increases it by one; interpreting the bit as an integer combines those cases in the given rule. Every nonempty string has a unique shorter prefix w and last bit x, so the recursion terminates and defines exactly one value for each finite string.

There is a unique decomposition of any nonempty bit string into its shorter prefix and its last bit. Removing the final bit reduces the length, and restoring it adds exactly its numerical value to the count of ones, so the recursion is well-defined.

I043

Recursion

2018 Final | Question 2(a) | Source page 4 | Difficulty 2
Q-2018-FINAL-P02-a-E3 | Answer provenance: official

> Let E_n be the number of ways in which you can tile a 3-by-n board with 1-by-3 pieces. For example, when $n = 4$ the board is 3-by-4 and $E_4 = 3$. The 3 ways are shown as follows:
>
> 
>
> (a) Calculate E_1, E_2, and E_3. [Atomic request: E_3.]

Official Answer

Source: physical PDF page 4 (2018-final.pdf).

1, 1, 2.

Supplementary Explanation

Answer. $E_3 = 2$.

Derivation/proof. If a vertical piece occurs, it fills a complete column. No horizontal piece can then cross all three columns, so the other columns must also be vertical. If no vertical piece occurs, each of the three rows must be one horizontal piece. These two disjoint cases are the all-vertical and all-horizontal tilings.

The three printed values are respectively E_1, E_2, E_3; this atomic question asks for the third.

On a three-column board the tiles are either all vertical or all horizontal. A horizontal tile forces the other two horizontal rows; otherwise all three columns are vertical, giving exactly two tilings.

I044

Recursion

2018 Final | Question 4(a) | Source page 6 | Difficulty 2
Q-2018-FINAL-P04-a | Answer provenance: official

> (a) Fill in the following recursive definition so that X is the set of all positive integers.
>
> Basis: $____ \in X$.
>
> Recursive step: If $a \in X$, then $a + 3 \in X$.
>
> For this question, no proof is required.

Official Answer

Source: physical PDF page 6 (2018-final.pdf).

$\{1, 2, 3\} \in X$.

Supplementary Explanation

Answer. Use basis elements $1, 2, 3 \in X$.

Derivation/proof. These three elements generate $1 + 3k, 2 + 3k, 3 + 3k$ for all integers $k \geq 0$, covering every positive integer by its remainder modulo three. The rule never produces a nonpositive value. At least one basis representative from each of the three residue classes is necessary; the source's braces mean three elements, not one set-valued element.

I045

Recursion

2018 Final | Question 4(b) | Source page 6 | Difficulty 2
Q-2018-FINAL-P04-b | Answer provenance: official

> (b) Fill in the following recursive definition so that $f(n) = \binom{n}{m}$, the binomial coefficient for fixed m.
>
> $f(m) = 1$
>
> $f(k + 1) = f(k) ____$
>
> For this question, no proof is required.

Official Answer

Source: physical PDF page 6 (2018-final.pdf).

$$f(k + 1) = f(k) \cdot \frac{k + 1}{k + 1 - m}.$$

Supplementary Explanation

Answer. $f(k + 1) = f(k)\frac{k+1}{k+1-m}$ for $k \geq m$.

Derivation/proof. Taking the ratio gives $\binom{k+1}{m} / \binom{k}{m} = [(k + 1)!/(m!(k + 1 - m)!)]/[k!/(m!(k - m)!)] = (k + 1)/(k + 1 - m)$. The denominator is positive for $k \geq m$, and the initial value $\binom{m}{m} = 1$ matches the given basis.

The recurrence multiplier is $\dfrac{k + 1}{k + 1 - m}$, obtained by cancelling factorials in successive binomial coefficients. The boundary value is $f(m) = \binom{m}{m} = 1$, so the rule begins without a zero denominator.

I046

Recursion

2018 Final | Question 4(c) | Source page 6 | Difficulty 2
Q-2018-FINAL-P04-c-edges | Answer provenance: official

> For this question, no proof is required. Fill in the following recursive definition so that S is the set of complete graphs.
>
> Recursive step: Suppose that $G = (V, E)$ is in S, where $V = \{v_1, v_2, \ldots, v_n\}$. Then $G' = (V', E')$ is in S, where $V' = \{v_1, v_2, \ldots, v_n, v_{n+1}\}$.
>
> (c) $E' =$ ____
>
> [Atomic request: edge set in the recursive step; no proof is required.]

Official Answer

Source: physical PDF page 6 (2018-final.pdf).

Recursive step: $E' = E \cup \{e'_1, e'_2, \ldots, e'_n\}$, where e'_i is the edge between vertex i and $n + 1$.

Supplementary Explanation

Answer. $E' = E \cup \{\{v_i, v_{n+1}\} : 1 \leq i \leq n\}$.

Derivation/proof. Every pair of old vertices is already an edge because G is complete. The only newly introduced pairs contain v_{n+1}, one with each of the n old vertices. Adding exactly those edges and retaining E supplies every distinct vertex pair once and adds no self-loops.

There are exactly n new edges, and the old-old and old-new pairs exhaust all distinct vertex pairs.

I047

Recursion

2017 Final | Question 10(a) | Source page 14 | Difficulty 2
Q-2017-FINAL-P10-a-second | Answer provenance: official

> Let S be the subset of the set of ordered pairs of integers defined recursively by:
>
> Basis step: $(0, 0) \in S$
>
> Recursive step: If $(a, b) \in S$, then $(a, b + 1) \in S$, $(a + 1, b + 1) \in S$, and $(a + 2, b + 1) \in S$.
>
> (a) List the elements of S produced by the first two applications of the recursive definition. [Atomic request: second application.]

Official Answer

Source: physical PDF page 14 (2017-final.pdf).

Second Application of Recursive step to S Omit the application of the Recursive step to $(0, 0)$ again, since we already have those terms in S. Applying recursive step to $(0, 1)$ gives $(0, 2), (1, 2), (2, 2)$. Applying recursive step to $(1, 1)$ gives $(1, 2), (2, 2), (3, 2)$. Applying recursive step to $(2, 1)$ gives $(2, 2), (3, 2), (4, 2)$. So $S = \{(0, 0), (0, 1), (0, 2), (1, 1), (1, 2), (2, 1), (2, 2), (3, 2), (4, 2)\}$.

Supplementary Explanation

Answer. The new elements are $(0, 2), (1, 2), (2, 2), (3, 2), (4, 2)$.

Derivation/proof. From $(0, 1)$ obtain first coordinates $0, 1, 2$ at level two; from $(1, 1)$ obtain $1, 2, 3$; from $(2, 1)$ obtain $2, 3, 4$. Their union gives $0, 1, 2, 3, 4$, each paired with 2. The accumulated set is these five pairs together with $(0, 0), (0, 1), (1, 1), (2, 1)$. Reapplying the rules to $(0, 0)$ adds no further elements.

The accumulated set has nine distinct elements.

After composing the possible intermediate steps, remove duplicates: a relation is a set, so an ordered pair reached by several intermediate elements is included only once.

I048

Recursion

2017 Final | Question 10(b) | Source page 14 | Difficulty 2 | 4 marks
Q-2017-FINAL-P10-b | Answer provenance: official

> Let S be the subset of the set of ordered pairs of integers defined recursively by:
>
> Basis step: $(0, 0) \in S$
>
> Recursive step: If $(a, b) \in S$, then $(a, b + 1) \in S$, $(a + 1, b + 1) \in S$, and $(a + 2, b + 1) \in S$.
>
> (b) Use structural induction to prove that $a \leq 2b$ whenever $(a, b) \in S$.

Official Answer

Source: physical PDF page 14 (2017-final.pdf).

Inductive Hypothesis: $P(a, b) : a \leq 2b$ whenever $(a, b) \in S$. Basis Step: $P(0, 0) : 0 \leq 0$. Recursive Step: $a \leq 2b$ (inductive hypothesis). $a \leq 2b \leq 2b + 2 \leq 2(b + 1)$. $P(a, b + 1)$ is true. $a \leq 2b$ (inductive hypothesis). $a + 1 \leq a + 2 \leq 2b + 2 \leq 2(b + 1)$. $P(a + 1, b + 1)$ is true. $a \leq 2b$ (inductive hypothesis). $a + 2 \leq 2b + 2 \leq 2(b + 1)$. $P(a + 2, b + 1)$ is true.

Supplementary Explanation

Answer. Every generated pair satisfies $a \leq 2b$.

Derivation/proof. The basis pair satisfies $0 \leq 2 \cdot 0$. Suppose (a, b) satisfies $a \leq 2b$. Every constructor produces $(a + j, b + 1)$ for $j \in \{0, 1, 2\}$. Then $a + j \leq 2b + 2 = 2(b + 1)$. Thus every constructor preserves the invariant. Since S contains only pairs finitely generated from the basis, structural induction proves the claim.

I049

Recursion

2017 Final | Question 11(a) | Source page 15 | Difficulty 2
Q-2017-FINAL-P11-a-T2 | Answer provenance: official

> In a Double Towers of Hanoi problem, there are three poles in a row and $2n$ disks for some positive integer n. There are two disks of size 1, two disks of size 2, and so on, up to two disks of size n. Initially one of the poles contains all the disks placed on top of each other in pairs of decreasing size. Disks are transferred one-by-one from one pole to another, but at no time may a larger disk be placed on top of a smaller disk. A disk may be placed on top of one of the same size. Let $T(n)$ be the number of moves needed to transfer a tower of $2n$ disks from one pole to another.
>
> (a) Find $T(1)$ and $T(2)$. [Atomic request: $T(2)$.]

Official Answer

Source: physical PDF page 15 (2017-final.pdf).

$T(2) = 6$.

Supplementary Explanation

Answer. $T(2) = 6$.

Derivation/proof. Move the two small disks to the spare pole in two moves, move the two large disks to the destination in two moves, and move the small pair onto them in two more moves. Before a large disk can move, both small disks must clear the source; afterward they must reach the destination. Thus at least two small-pair transfers plus two large-disk moves are necessary, totaling six.

The three pair transfers show $T(2) \leq 2 + 2 + 2 = 6$. For the lower bound, before the first large disk moves the two small disks must leave its pole; two large-disk moves and a final transfer of the small pair are also unavoidable.

I050

Recursion

2017 Final | Question 11(b) | Source page 15 | Difficulty 2 | 2 marks
Q-2017-FINAL-P11-b | Answer provenance: official

> In a Double Towers of Hanoi problem, there are three poles in a row and $2n$ disks for some positive integer n. There are two disks of size 1, two disks of size 2, and so on, up to two disks of size n. Initially one of the poles contains all the disks placed on top of each other in pairs of decreasing size. Disks are transferred one-by-one from one pole to another, but at no time may a larger disk be placed on top of a smaller disk. A disk may be placed on top of one of the same size. Let $T(n)$ be the number of moves needed to transfer a tower of $2n$ disks from one pole to another.
>
> (b) Find $T(3)$.

Official Answer

Source: physical PDF page 15 (2017-final.pdf).

$T(3) = 14$.

Supplementary Explanation

Answer. $T(3) = 14$.

Derivation/proof. The top two size-pairs form a smaller instance taking $T(2) = 6$ moves. Move that subtower to the spare pole, move the largest equal pair to the destination in two moves, then transfer the subtower onto it in another six. This gives $6 + 2 + 6 = 14$, and each of these phases is necessary because no larger disk can rest on a smaller disk.

I051

Recursion

2016 Final | Question 7(a) | Source page 13 | Difficulty 2
Q-2016-FINAL-P07-a-recurrence | Answer provenance: official

> Assume that a party of n people wants to select some of their members to form a committee. The committee must have at least one member. Let $T(n)$ be the total number of possible committees that can be formed from a party of n people.
>
> (a) Define $T(n)$ as a recurrence equation, and prove that your recurrence equation is correct by induction.
>
> Atomic focus [editorial]: define the recurrence; its induction proof is a separate atom.

Official Answer

Source: PDF physical page 13 (2016-final.pdf).

(a) $T(1) = 1$, and $T(n) = 2T(n-1) + 1$ for $n > 1$.

The base case $T(1)$ is obviously true.

Assume that $T(n-1)$ counts the number of committees that are formed from $n-1$ people correctly for $n > 1$.

When there are n people, the number of possible committees that are formed without the n-th person is counted by $T(n-1)$. On the other hand, each committee without the n-th person union with the n-th person forms another possible committee. So, we have $2T(n-1)$ possible committees. There is only one case is missed, which is the committee that contains only the n-th person. Therefore, when there are n people, $T(n) = 2T(n-1) + 1$ counts all possible committees with at least one member from n people. By the principle of induction, we have proved that the recurrence equation is correct.

Supplementary Explanation

Answer. $T(1) = 1$ and $T(n) = 2T(n-1) + 1$ for $n > 1$.

Derivation/proof. With one person there is exactly one allowed committee. For $n > 1$, old nonempty committees contribute $T(n-1)$ without person n and another $T(n-1)$ with person n added. The remaining allowed committee is $\{n\}$, giving the recurrence. These cases are disjoint and exhaustive.

I052

Recursion

2016 Final | Question 7(b) | Source page 13 | Difficulty 2 | 3 marks
Q-2016-FINAL-P07-b | Answer provenance: official

> Assume that a party of n people wants to select some of their members to form a committee. The committee must have at least one member. Let $T(n)$ be the total number of possible committees that can be formed from a party of n people.
>
> (b) Iterate the recurrence equation $T(n)$ in part (a) to obtain a closed form solution.

Official Answer

Source: PDF physical page 13 (2016-final.pdf).

(b)

$$\begin{aligned} T(n) &= 2T(n-1)+1 \\ &= 2^2T(n-2)+2+1 \\ &= 2^3T(n-3)+2^2+2+1 \\ &\;\;\vdots \\ &= 2^{n-1}T(1)+2^{n-2}+\cdots+2^1+2^0 \\ &= 2^{n-1}+2^{n-2}+\cdots+2^1+2^0 \\ &= 2^n-1 \end{aligned}$$

Supplementary Explanation

Answer. $T(n)=2^n-1$.

Derivation/proof. Starting with $T(n)=2T(n-1)+1$, after j expansions, $T(n)=2^jT(n-j)+\sum_{i=0}^{j-1}2^i$. Set $j=n-1$: $T(n)=2^{n-1}T(1)+\sum_{i=0}^{n-2}2^i=2^{n-1}+(2^{n-1}-1)=2^n-1$. For $n=1$, the sum is empty and the same formula gives 1.

I053

Recursion

2015 Final | Question 8(a) | Source page 14 | Difficulty 2 | 4 marks
Q-2015-FINAL-P08-a | Answer provenance: official

(a) Consider the recurrence below defined for $n \geq 0$.

$$T(n) = \begin{cases} 3 & \text{if } n = 0, \\ 7T(n-1) + 12 & \text{if } n > 0. \end{cases}$$

Give a closed-form solution to the recurrence. You only have to give the solution. You do not need to show how you derived it.

Official Answer

Source: PDF physical page 14 (2015-final.pdf).

(a) By iterating the recurrence, we get:

$$T(n) = 3 \cdot 7^n + 12 \cdot \frac{7^n - 1}{7 - 1} = 5 \cdot 7^n - 2.$$

Supplementary Explanation

Answer. $T(n) = 5 \cdot 7^n - 2$.

Derivation/proof. Iteration yields $T(n) = 3 \cdot 7^n + 12 \sum_{j=0}^{n-1} 7^j = 3 \cdot 7^n + 12(7^n - 1)/6 = 5 \cdot 7^n - 2$. At $n = 0$, the formula gives 3, matching the initial condition.

I054

Recursion

2015 Final | Question 9 | Source page 15 | Difficulty 2 | 3 marks
Q-2015-FINAL-P09-recurrence | Answer provenance: official

> Let A and B be two positive integers represented as binary sequences, each n bits long. Assume n is a power of 2. The product $A \cdot B$ can be computed recursively as follows:
>
> 1. If $n = 1$, calculate $A \cdot B$ directly.
> 2. If $n > 1$, partition the bit representation of each number into two parts, the high $n/2$ bits and low $n/2$ bits. Calculate $A \cdot B$ using the following formula:
>
> $$\begin{aligned} A \cdot B &= (A_h \cdot 1\underbrace{0 \cdots 0}_{n/2} + A_l) \cdot (B_h \cdot 1\underbrace{0 \cdots 0}_{n/2} + B_l) \\ &= (A_h \cdot B_h) \cdot 1\underbrace{0 \cdots 0}_{n} + (A_h \cdot B_l) \cdot 1\underbrace{0 \cdots 0}_{n/2} \\ &\quad + (A_l \cdot B_h) \cdot 1\underbrace{0 \cdots 0}_{n/2} + (A_l \cdot B_l). \end{aligned} \tag{2}$$
>
> where A_h and B_h denote the high $n/2$ bits of A and B, respectively, and A_l and B_l denote the low $n/2$ bits of A and B, respectively. The terms in ( ) are calculated recursively.
>
> Suppose the calculations take 1 unit of time when $n = 1$, and suppose that calculation of the expression (2) takes cn units of time given the results of the recursive calls, where c is a constant.
>
> Let $T(n)$ be the total time it takes to compute $A \cdot B$. Write a recurrence for $T(n)$, find a closed-form expression for it, and simplify the closed-form expression using the big Θ notation.
>
> Write a recurrence for $T(n)$,

Official Answer

Source: PDF physical page 16 (2015-final.pdf).

According to the above expression of $A \cdot B$,

$$T(n) = 4T(\frac{n}{2}) + cn$$

Supplementary Explanation

Answer. $T(1) = 1, T(n) = 4T(n/2) + cn$ for powers of two $n > 1$.

Derivation/proof. Each of the four products in the expansion takes $T(n/2)$ time. Shifts and additions after these results are available cost cn by hypothesis. Therefore their sum is $4T(n/2) + cn$, with the specified direct-multiplication base case.

The four products are $A_hB_h, A_hB_l, A_lB_h, A_lB_l$. The base $T(1) = 1$ is direct multiplication, with no additional combine term.

I055

Recursion

2025 Fall Final | Question 10(ii) | Source page 17 | Difficulty 3 | 4 marks
Q-2025F-FINAL-P10-ii | Answer provenance: official

> A spider starts with 10 legs. Each day, its legs instantly double by splitting, but immediately after, exactly 2 legs fall off and are lost forever. Let a_n be the number of legs after n days, with $a_0 = 10$.
>
> (ii) [4 points] Find the closed-form formula for a_n using the expansion method.

Official Answer

Source: physical PDF page 17 (COMP2711_F2025_Final.pdf).

(ii) We use the recurrence $a_n = 2a_{n-1} - 2$ and substitute repeatedly:

$$\begin{aligned} a_n &= 2a_{n-1} - 2 \\ &= 2(2a_{n-2} - 2) - 2 = 2^2 a_{n-2} - 2 \cdot 2 - 2 \\ &= 2^2(2a_{n-3} - 2) - 2 \cdot 2 - 2 = 2^3 a_{n-3} - 2 \cdot 2^2 - 2 \cdot 2 - 2 \\ &\vdots \\ &= 2^n a_0 - 2(2^{n-1} + 2^{n-2} + \cdots + 2 + 1). \end{aligned}$$

The sum in parentheses is a geometric series:

$$2^{n-1} + 2^{n-2} + \cdots + 2 + 1 = 2^n - 1.$$

Thus,

$$a_n = 2^n a_0 - 2(2^n - 1) = 2^n \cdot 10 - 2 \cdot 2^n + 2 = 10 \cdot 2^n - 2^{n+1} + 2.$$

Simplifying:

$$a_n = 10 \cdot 2^n - 2^{n+1} + 2 = (10 - 2) \cdot 2^n + 2 = 8 \cdot 2^n + 2.$$

Therefore the explicit formula is

$$a_n = 8 \cdot 2^n + 2 \quad \text{for all } n \geq 0.$$

Supplementary Explanation

Answer. $a_n = 8 \cdot 2^n + 2$.

Derivation/proof. Start with $a_n = 2a_{n-1} - 2$ and $a_0 = 10$. Expanding twice gives $a_n = 2^2 a_{n-2} - 2(2 + 1)$; three expansions give $2^3 a_{n-3} - 2(2^2 + 2 + 1)$. Continue to day zero:

$$a_n = 2^n a_0 - 2 \sum_{j=0}^{n-1} 2^j.$$

The finite geometric sum equals $2^n - 1$, so

$$a_n = 10 \cdot 2^n - 2(2^n - 1) = 8 \cdot 2^n + 2.$$

For $n = 0$, the empty sum is zero and the formula gives ten. For $n = 1$ it gives 18, consistent with the first day's update.

I056

Recursion

2024 Spring Final | Question 5 | Source page 6 | Difficulty 3 | 10 marks
Q-2024S-FINAL-P05 | Answer provenance: official

> You have a set of n biased coins. For each $m = 1, 2, \ldots, n$, the m-th coin has a probability of $\frac{1}{2m+1}$ of landing heads when tossed, and the outcomes of the tosses for each coin are independent of each other. Show that the probability of obtaining an odd number of heads after tossing all the n coins is equal to $\frac{n}{2n+1}$. Hint: One possible approach is to think in a recursive way.

Official Answer

Source: physical PDF page 6 (COMP2711_S2024_Final.pdf).

> Solution: Let P_n be the probability of obtaining an odd number of heads when tossing these n coins. In order to compute P_{n+1}, we need to consider two cases: Case 1: The $(n+1)$-th coin lands head and we have an even number of heads in the first n coins. Case 2: The $(n+1)$-th coin lands tail and we have an odd number of heads in the first n coins. Thus:
>
> $$P_{n+1} = \left(\frac{1}{2n+3}\right)(1 - P_n) + \left(1 - \frac{1}{2n+3}\right)P_n = \left(1 - \frac{2}{2n+3}\right)P_n + \frac{1}{2n+3}.$$
>
> Multiplying both sides by $(2n+3)$ we obtain $(2n+3)P_{n+1} = (2n+1)P_n + 1$. Substituting $A_n = (2n+1)P_n$, we can express this as $A_{n+1} = A_n + 1$. We know that $A_0 = 0$ because $P_0 = 0$. This implies that $A_n = n$, which in turn implies that $P_n = \frac{n}{2n+1}$, as desired.

Supplementary Explanation

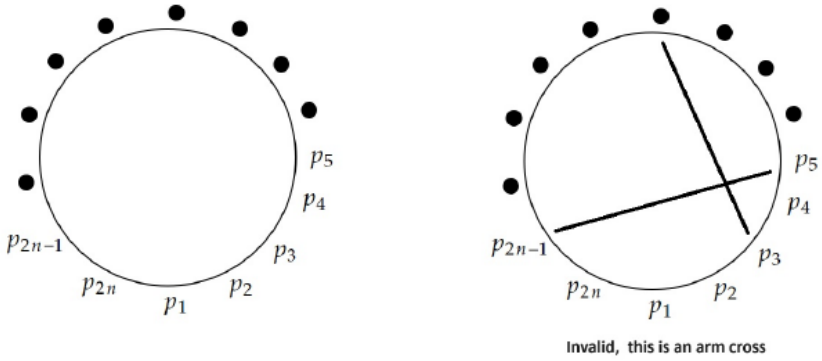
Answer. $P_n = n/(2n+1)$.

Derivation/proof. Let P_n denote the odd-head probability for the first n coins, with $P_0 = 0$. Coin $n+1$ has head probability $1/(2n+3)$. An odd total comes from an even previous total and a head, or an odd previous total and a tail. These cases are mutually exclusive, and independence gives $P_{n+1} = \frac{1}{2n+3}(1 - P_n) + \frac{2n+2}{2n+3}P_n$. Hence $(2n+3)P_{n+1} = (2n+1)P_n + 1$. Set $A_n = (2n+1)P_n$. Then $A_{n+1} = A_n + 1$, $A_0 = 0$, so $A_n = n$ and $P_n = n/(2n+1)$. At $n = 1$ this gives $1/3$, as required.

I057

Recursion

2023 Fall Final | Question 9(c) | Source page 14 | Difficulty 3 | 5 marks
Q-2023F-FINAL-P09-c | Answer provenance: official

> Assume that there are $2n$ people sitting around a round table. We want to split them into n pairs to engage in handshakes so that no arms cross, given that nobody is allowed to shake hands with more than one person at once. Let a_{2n} denote the number of ways that we can pair them legally without any arm crossing. You can see an invalid handshake in the figure below. Please answer the following questions by referring to the figure:
>
> 
>
>
> Assume p_1 is paired with p_j initially. Derive the number of ways to pair up the remaining people in terms of a_i.
>
> (Hint: Notice that this handshake will create two separate groups of people on the “left” side and the “right” side of it)

Official Answer

Source: physical PDF page 14 (COMP2711_F2023_Final.pdf).

> (c) In this case we have one group of size $j-2$ on one side and a group of size $2n-j$ on the other side. So We have $a_{j-2} \cdot a_{2n-j}$ number of ways to pair the remaining people in a valid way.

Supplementary Explanation

Answer. $a_{j-2}a_{2n-j}$ for even $j \in \{2, 4, \ldots, 2n\}$; zero for odd j.

Derivation/proof. A chord from p_1 to p_j separates $p_2, \ldots, p_{j-1}$ from $p_{j+1}, \ldots, p_{2n}$. The groups have sizes $j-2$ and $2n-j$. If j is odd, both are odd and no complete internal pairing exists. For even j, there are a_{j-2} noncrossing pairings in the first group and a_{2n-j} in the second. Any pair of such choices combines with the fixed chord uniquely and introduces no crossings between regions. Conversely, every legal pairing containing the fixed chord restricts to those two choices. This bijection proves the product.

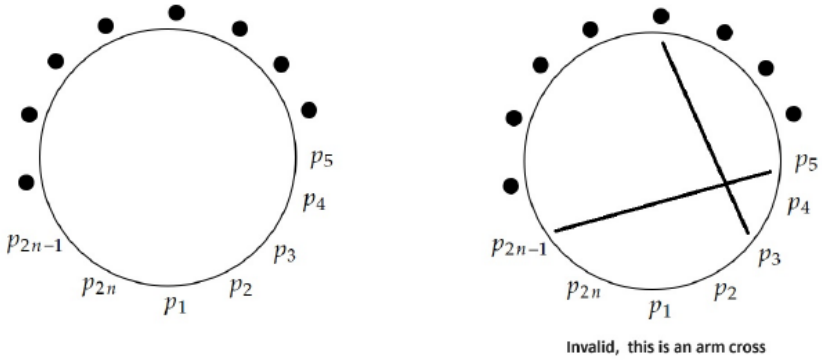
The count is $a_{j-2}a_{2n-j}$ for even j, and is zero for odd j. There is a bijection between completions and ordered pairs of valid side matchings; an empty side contributes its unique empty matching, counted by $a_0 = 1$.

I058

Recursion

2023 Fall Final | Question 9(d) | Source page 14 | Difficulty 3 | 3 marks
Q-2023F-FINAL-P09-d | Answer provenance: official

Assume that there are $2n$ people sitting around a round table. We want to split them into n pairs to engage in handshakes so that no arms cross, given that nobody is allowed to shake hands with more than one person at once. Let a_{2n} denote the number of ways that we can pair them legally without any arm crossing. You can see an invalid handshake in the figure below. Please answer the following questions by referring to the figure:



Show the recursive relation of a_{2n}. You do not have to represent it in closed form. You can assume $a_0 = 1$.

Official Answer

Source: physical PDF page 14 (COMP2711_F2023_Final.pdf).

> (d) We just need to calculate the summation in which the first person is paired with $p_2, p_4, p_6, \cdots, p_{2n}$.

Supplementary Explanation

Answer. $a_0 = 1$, and $a_{2n} = \sum_{k=1}^{n} a_{2k-2}a_{2n-2k}$ for $n \geq 1$.

Derivation/proof. The feasible partners of p_1 are exactly p_{2k} with $1 \leq k \leq n$. For this fixed partner, the two blocks have $2k-2$ and $2n-2k$ people, so part (c) supplies the product $a_{2k-2}a_{2n-2k}$. Summing these disjoint and exhaustive classes gives the recurrence. The empty pairing counts as one, so $a_0 = 1$. Checks: $a_2 = 1$, $a_4 = 1+1 = 2$, and $a_6 = 2+1+2 = 5$. No closed form is needed.

The recurrence is $a_{2n} = \sum_{k=1}^{n} a_{2k-2}a_{2n-2k}$ with $a_0 = 1$. It gives $a_2 = 1$, $a_4 = 2$, and $a_6 = 2+1+2 = 5$.

I059

Recursion

2021 Final Part 1 | Question 2 | Source page 1 | Difficulty 3 | 10 marks
Q-2021-FINAL1-P02 | Answer provenance: official

> There are n coins, all of equal shape and size, but only $n-1$ of them are of equal weight. You are given a balance scale which can compare the weight of any two sets of coins. Assume n is a power of 3. We use the following algorithm to find the coin of different weight.
>
> ```
> Procedure findCoin(S : a set of coins)
>     if |S| = 1:
>         return the only coin in S.
>     Partition S equally into 3 sets of coins, A, B and C.
>     if A and B are of equal weight:
>         return findCoin(C)
>     else if B and C are of equal weight:
>         return findCoin(A)
>     else:
>         return findCoin(B).
> ```
>
> Show that the number of weightings is at most $2\log_3 n$.

Official Answer

Source: physical PDF page 2 (2021-final-part-1.pdf).

> In each recursive call, it takes at most 2 comparisons and the input size are reduced to one-third of the parent. So, the number of comparisons is at most $2\log_3 n = O(\log n)$.

Supplementary Explanation

Answer. At most $2\log_3 n$ weighings.

Derivation/proof. At a nontrivial call, at most two balance comparisons determine the unique group containing the odd coin. Equal A,B means C contains it; unequal A,B and equal B,C means A; if both comparisons differ, B contains it. Only that group, of size $|S|/3$, is recursed into. For n=3^k there are exactly k nontrivial size levels before size one, whose case uses no weighing. Hence total comparisons ≤2k=2log3 n. This also establishes why only one branch is followed.

If both comparisons differ, the odd coin is in B: an odd coin in A would leave B, C balanced, and an odd coin in C would leave A, B balanced. This reasoning does not require knowing whether the coin is heavier or lighter. The successive sizes are $3^k, 3^{k-1}, \ldots, 3, 1$.

Thus the weighing bound is $2k = 2\log_3 n$.

I060

Recursion

2021 Final Part 1 | Question 6 | Source page 3 | Difficulty 3 | 10 marks
Q-2021-FINAL1-P06 | Answer provenance: official

> A string is called E-String if the string consists of only digits in $\{0, 2, 4, 7, 8, 9\}$ and the sum of the digits is even. E.g. "079" is an E-String because $0 + 7 + 9 = 16$ is even. "708" is not an E-String because $7 + 0 + 8 = 15$ is odd. "107" is not an E-String because the digit 1 is not allowed. The empty string is also considered an E-string.
>
> How many E-Strings of length n are there? Give a closed-form solution. [Hint: use a recurrence.]

Official Answer

Source: physical PDF page 4 (2021-final-part-1.pdf).

> Let $E(n)$ denote the number of E-Strings of length n.
>
> **Base case:** $E(0) = 1$
>
> **Recursive case:** For $n > 0$, $E(n) = 2 \cdot (6^{n-1} - E(n-1)) + 4 \cdot E(n-1) = 2 \cdot 6^{n-1} + 2 \cdot E(n-1)$.
>
> $$\begin{aligned} E(n) &= 2 \cdot 6^{n-1} + 2 \cdot E(n-1) \\ &= 2 \cdot 6^{n-1} + 2 \cdot (2 \cdot 6^{n-2} + 2 \cdot E(n-2)) \\ &= 2 \cdot 6^{n-1} + 2^2 \cdot 6^{n-2} + 2^2 \cdot E(n-2) \\ &\vdots \\ &= \sum_{i=1}^{n} 2^i \cdot 6^{n-i} + 2^n \cdot E(0) \\ &= \sum_{i=1}^{n} 2^n \cdot 3^{n-i} + 2^n \\ &= 2^n \sum_{i=0}^{n-1} 3^i + 2^n \\ &= 2^n \frac{3^n - 1}{3 - 1} + 2^n \\ &= 2^{n-1}(3^n - 1) + 2^n. \end{aligned}$$

Supplementary Explanation

Answer. $E_n = (6^n + 2^n)/2$.

Derivation/proof. There are four even digits and two odd digits. With E0=1, append an even digit to an even-sum prefix (4E_(n−1) choices) or an odd digit to an odd-sum prefix (2(6^(n−1)−E_(n−1)) choices). Thus E_n=2E_(n−1)+2·6^(n−1). The proposed closed form equals 1 at n=0. Substitution gives 2[(6^(n−1)+2^(n−1))/2]+2·6^(n−1)=3·6^(n−1)+2^(n−1)=(6^n+2^n)/2. This proves it by recurrence induction; E1=4 also checks the alphabet count.

The even digits are $0, 2, 4, 8$ and the odd digits are $7, 9$. Explicitly,

$$E_n = 4E_{n-1} + 2(6^{n-1} - E_{n-1}) = 2E_{n-1} + 2 \cdot 6^{n-1}.$$

Unrolling gives

$$E_n = 2^n E_0 + \sum_{i=1}^{n} 2^i 6^{n-i} = 2^n + 2^n \sum_{j=0}^{n-1} 3^j = 2^n + 2^n(3^n - 1)/2 = (6^n + 2^n)/2.$$

The final answer contains no summation, as required by the paper.

As required, the final answer contains no summation.

I061

Recursion

2020 Final Part 1 | Question 5(b) | Source page 3 | Difficulty 3 | 5 marks
Q-2020-FINAL1-P05-b | Answer provenance: official

> Consider the following series of functions defined recursively: $f_0(x) = x^{-1}$; for $n \geq 0$, $f_{n+1}(x) = f_n'(x)$ (recall that $(ax^b)' = abx^{b-1}$).
>
> Let $a_n = \log(-f_n(-1))$. Which one of the following functions is the tight asymptotic bound (i.e., Θ) of a_n? Justify your answer.
>
> $n,\ n \log\log n,\ n \log n,\ n \log^2 n,\ n^2$

Official Answer

Source: physical PDF page 4 (2020-final-part-1.pdf).

$a_n = \log n! = \sum_{i=1}^{n} \log i$. On the one hand, $a_n < \sum_{i=1}^{n} \log n = n \log n$, so $a_n = O(n \log n)$. On the other hand, $a_n > \sum_{i=n/2+1}^{n} \log(n/2) = \frac{n}{2} \log \frac{n}{2}$, so $a_n = \Omega(n \log n)$. Therefore we conclude $a_n = \Theta(n \log n)$, and the tightest asymptotic upper bound is $n \log n$.

Supplementary Explanation

Answer. $n \log n$; $a_n = \Theta(n \log n)$.

Derivation/proof. The formula gives $f_n(-1) = (-1)^n n!(-1)^{-(n+1)} = -n!$, hence $a_n = \log(n!)$. For n≥4, all n factors of n! are at most n, so $a_n \leq n \log n$. At least n/2 factors are at least n/2, giving $a_n \geq (n/2)\log(n/2) \geq (n/4)\log n$. These fixed positive upper and lower bounds prove the claimed Θ order without needing Stirling's formula.

The constants $1/4$ and 1 with threshold 4 work for odd as well as even n and need no Stirling formula.

Substitution first gives $f_n(-1) = -n!$, so $\log(-f_n(-1)) = \log(n!)$ is defined. The lower and upper constants can be chosen as $1/4, 1$ respectively for $n \geq 4$, proving the two-sided asymptotic estimate.

I062

Recursion

2020 Final Part 1 | Question 6(b) | Source page 5 | Difficulty 3 | 4 marks
Q-2020-FINAL1-P06-b | Answer provenance: official

Read the following algorithms and answer the questions.

Algorithm 1: MultiplyTwoPolys($p(x), q(x)$) Input: A polynomial $p(x)$ with degree m and a polynomial $q(x)$ with degree n Output: The polynomial $r(x) = p(x) \cdot q(x)$

```
for i ← 0 to m do
    a_i ← the coefficient of x^i in p(x)
end
for i ← 0 to n do
    b_i ← the coefficient of x^i in q(x)
end
for i ← 0 to m+n do
    c_i ← 0
end
for i ← 0 to m do
    for j ← 0 to n do
        c_(i+j) ← c_(i+j) + a_i b_j
    end
end
r(x) ← sum from i=0 to m+n of c_i x^i
return r(x)
```

Algorithm 2: MultiplyManyPolys($p_1(x), p_2(x), \ldots, p_n(x)$) Input: Polynomials $p_1(x), p_2(x), \ldots, p_n(x)$ Output: The polynomial $r(x) = p_1(x)p_2(x)\cdots p_n(x)$

```
if n = 1 then
    return p_1(x)
end
r_1(x) ← MultiplyManyPolys(p_1(x),p_2(x),…,p_(n−1)(x))
r(x) ← MultiplyTwoPolys(r_1(x),p_n(x))
return r(x)
```

Let $\{p_n(x)\}$ be a sequence of polynomials, all having degree 1. What is the time complexity of computing $p_1(x)p_2(x)\cdots p_n(x)$ using Algorithm 2? Write your answer in the best big-O notation with respect to n. Justify your answer.

Official Answer

Source: physical PDF page 5 (2020-final-part-1.pdf).

Let C_n be the time complexity of computing $p_1(x)p_2(x)\cdots p_n(x)$ using Algorithm 2, then $C_1 = 0$ and $C_n = C_{n-1} + n - 1$. Therefore $C_n = \sum_{i=1}^{n}(i-1) = O(n^2)$.

Supplementary Explanation

Answer. $O(n^2)$, in fact $\Theta(n^2)$.

Derivation/proof. A product of k−1 degree-one polynomials has degree k−1. Multiplying it by the kth degree-one polynomial with Algorithm 1 uses Θ(k) coefficient operations. Algorithm 2 does this sequentially for k=2,…,n, so $T(n) = T(n-1) + \Theta(n)$ with T(1)=Θ(1). Expanding sums the arithmetic sequence, yielding Θ(n²). The recursion has only one child at each step, not a balanced multiplication tree.

The last level performs exactly $2n$ coefficient multiplications, and the total for levels two through n is twice the arithmetic sum $2 + 3 + \cdots + n = n(n+1)/2 - 1$. This quadratic count yields $T(n) = \Theta(n^2)$; the displayed recurrence is an asymptotic simplification rather than the exact loop count.

I063

Recursion

2019 Final | Question 6(b) | Source page 8 | Difficulty 3 | 6 marks
Q-2019-FINAL-P06-b | Answer provenance: official

> (a) Give a recursive definition of the function $ones(s)$, which counts the number of ones in a bit string s.
>
> (b) Use structural induction to prove that $ones(st) = ones(s) + ones(t)$.

(b) Use structural induction to prove that $ones(st) = ones(s) + ones(t)$.

Official Answer

Source: physical PDF page 8 (2019-final.pdf).

We prove by induction on the length of t. The basis step is when $t = \lambda$, in which case we have $\text{ones}(s\lambda) = \text{ones}(s) = \text{ones}(s) + 0 = \text{ones}(s) + \text{ones}(\lambda)$. For the inductive step, write $t = wx$, where w is a bit string and x is a bit. Then we have

$$\begin{aligned}
\text{ones}(s(wx)) &= \text{ones}((sw)x) \\
&= x + \text{ones}(sw) && \text{(by definition)} \\
&= x + \text{ones}(s) + \text{ones}(w) && \text{(by the inductive hypothesis)} \\
&= \text{ones}(s) + (x + \text{ones}(w)) \\
&= \text{ones}(s) + \text{ones}(wx) && \text{(by definition)} \\
&= \text{ones}(s) + \text{ones}(t).
\end{aligned}$$

Supplementary Explanation

Answer. The identity holds for all finite bit strings s,t.

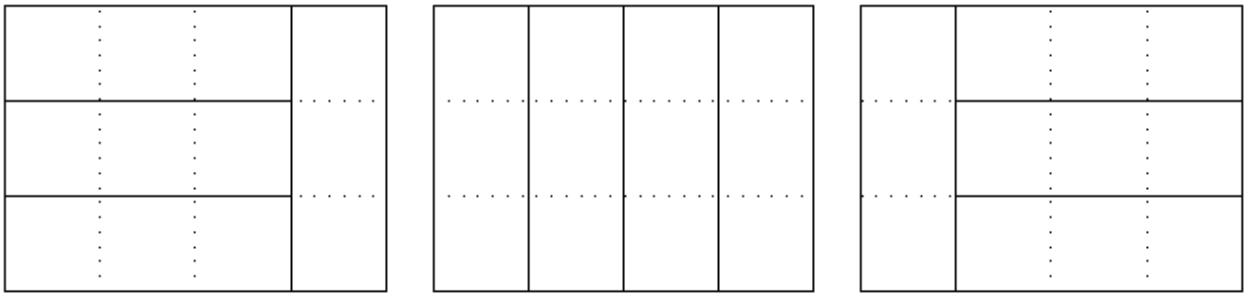
Derivation/proof. Fix any s and induct structurally on t. For $t = \lambda$, $s\lambda = s$ and $\text{ones}(\lambda) = 0$, proving the base. Assume the identity for a prefix w and append a bit x. Then $\text{ones}(s(wx)) = \text{ones}((sw)x) = \text{ones}(sw) + x = \text{ones}(s) + \text{ones}(w) + x = \text{ones}(s) + \text{ones}(wx)$. The middle equality uses the induction hypothesis for w; the other count equalities use the recursive definition. Thus the constructor preserves the identity.

For the structural-induction basis, λ is the empty string and $\text{ones}(s\lambda) = \text{ones}(s) = \text{ones}(s) + \text{ones}(\lambda)$. Appending the empty string changes neither the string nor the number of ones.

I064

Recursion

2018 Final | Question 2(b) | Source page 4 | Difficulty 3
Q-2018-FINAL-P02-b | Answer provenance: official

> Let E_n be the number of ways in which you can tile a 3-by-n board with 1-by-3 pieces. For example, when $n = 4$ the board is 3-by-4 and $E_4 = 3$. The 3 ways are shown as follows:
>
> 
>
> (b) Show that $E_n = E_{n-1} + E_{n-3}$. (Hint: You may use a counting argument.)

Official Answer

Source: physical PDF page 4 (2018-final.pdf).

Consider the top-rightmost tile of a 3-by-n board. If it is vertical, the remaining board is 3-by-$(n-1)$, so there are E_{n-1} ways to tile it. If it is horizontal, it must be followed by two more horizontal tiles below it, so there are E_{n-3} ways to tile it. Therefore $E_n = E_{n-1} + E_{n-3}$.

Supplementary Explanation

Answer. $E_n = E_{n-1} + E_{n-3}$ for $n \geq 4$ (also $n = 3$ with $E_0 = 1$).

Derivation/proof. Examine the tile covering the top-right cell. If vertical, it occupies the last column, leaving a 3-by-$(n-1)$ board. If horizontal, it covers the top cells of the last three columns. Each remaining cell in the rightmost column must then be covered horizontally: a vertical piece would overlap that top tile. Those two horizontal pieces also end at the boundary and fill the same three columns. This leaves a 3-by-$(n-3)$ board. Removing or adding the forced boundary tiles gives bijections to the two smaller tiling sets; the cases are disjoint and exhaustive.

With only positive-index initial values, use the recurrence for $n \geq 4$. Defining the empty-board count $E_0 = 1$ also makes it valid at $n = 3$.

Separate tilings according to their right boundary. A vertical last tile leaves one fewer column, whereas horizontal tiles occupy the rightmost three columns. Removing and restoring these forced tiles is a one-to-one correspondence with the two smaller-board families.

I065

Recursion

2017 Final | Question 11(c) | Source page 15 | Difficulty 3 | 3 marks
Q-2017-FINAL-P11-c | Answer provenance: official

> In a Double Towers of Hanoi problem, there are three poles in a row and $2n$ disks for some positive integer n. There are two disks of size 1, two disks of size 2, and so on, up to two disks of size n. Initially one of the poles contains all the disks placed on top of each other in pairs of decreasing size. Disks are transferred one-by-one from one pole to another, but at no time may a larger disk be placed on top of a smaller disk. A disk may be placed on top of one of the same size. Let $T(n)$ be the number of moves needed to transfer a tower of $2n$ disks from one pole to another.
>
> (c) Find a recurrence relation expressing $T(n)$ in terms of $T(n-1)$, for all integers $n \geq 2$.

Official Answer

Source: physical PDF page 15 (2017-final.pdf).

$T(n) = 2T(n-1) + 2$.

Supplementary Explanation

Answer. $T(n) = 2T(n-1) + 2$, with $T(1) = 2$.

Derivation/proof. To expose the largest pair, first transfer all $2(n-1)$ smaller disks to the spare pole in $T(n-1)$ moves. Transfer the two largest disks directly in two moves, then transfer the smaller tower onto them in $T(n-1)$ moves. Conversely, before the first largest disk moves all smaller disks must have cleared its source, and after the last largest disk reaches the destination the smaller tower must be transferred onto it. These disjoint phases need at least those two subtower transfers and two largest-disk moves, so the count is exact.

Before the first largest disk moves, neither its source nor target can contain smaller disks, so every smaller disk lies on the third pole. After the last largest disk reaches the destination, the smaller tower lies on another pole and must still be transferred. These before-and-after phases are disjoint.

The essential lower-bound intervals are before the first and after the last largest-disk move. Each interval requires transferring the whole stack of smaller pairs, while the two largest disks themselves require two moves, so no shorter route avoids either recursive cost.

I066

Recursion

2017 Final | Question 11(d) | Source page 15 | Difficulty 3 | 3 marks
Q-2017-FINAL-P11-d | Answer provenance: official

> In a Double Towers of Hanoi problem, there are three poles in a row and $2n$ disks for some positive integer n. There are two disks of size 1, two disks of size 2, and so on, up to two disks of size n. Initially one of the poles contains all the disks placed on top of each other in pairs of decreasing size. Disks are transferred one-by-one from one pole to another, but at no time may a larger disk be placed on top of a smaller disk. A disk may be placed on top of one of the same size. Let $T(n)$ be the number of moves needed to transfer a tower of $2n$ disks from one pole to another.
>
> (d) Find a closed-form solution for the recurrence.

Official Answer

Source: physical PDF page 15 (2017-final.pdf).

$$\begin{aligned} T(n) &= 2T(n-1)+2 \\ &= 2(2T(n-2)+2)+2 \\ &= 2^2T(n-2)+2^2+2 \\ &\quad \cdots \\ &= 2^{n-1}T(n-(n-1))+2^{n-1}+\cdots+2^2+2^1 \\ &= 2^{n-1}T(1)+2^{n-1}+\cdots+2^2+2^1 \\ &= 2^n+2^{n-1}+\cdots+2^2+2^1 \\ &= 2^{n+1}-2. \end{aligned}$$

Supplementary Explanation

Answer. $T(n)=2^{n+1}-2$.

Derivation/proof. From $T(n)=2T(n-1)+2$, set $U(n)=T(n)+2$. Then $U(n)=2U(n-1)$ and $U(1)=4$, hence $U(n)=4\cdot 2^{n-1}=2^{n+1}$. Subtract two. Independently, $n=1$ gives 2, and substitution gives $2(2^n-2)+2=2^{n+1}-2$, verifying the initial value and recurrence for all $n\geq 2$.

The initial value and recurrence uniquely determine the sequence.

I067

Recursion

2016 Final | Question 7(a) | Source page 13 | Difficulty 3
Q-2016-FINAL-P07-a-proof | Answer provenance: official

> Assume that a party of n people wants to select some of their members to form a committee. The committee must have at least one member. Let $T(n)$ be the total number of possible committees that can be formed from a party of n people.
>
> (a) Define $T(n)$ as a recurrence equation, and prove that your recurrence equation is correct by induction.
>
> Atomic focus [editorial]: prove the recurrence counts committees; defining it is a separate atom.

Official Answer

Source: PDF physical page 13 (2016-final.pdf).

(a) $T(1) = 1$, and $T(n) = 2T(n-1) + 1$ for $n > 1$.

The base case $T(1)$ is obviously true.

Assume that $T(n-1)$ counts the number of committees that are formed from $n-1$ people correctly for $n > 1$.

When there are n people, the number of possible committees that are formed without the n-th person is counted by $T(n-1)$. On the other hand, each committee without the n-th person union with the n-th person forms another possible committee. So, we have $2T(n-1)$ possible committees. There is only one case is missed, which is the committee that contains only the n-th person. Therefore, when there are n people, $T(n) = 2T(n-1) + 1$ counts all possible committees with at least one member from n people. By the principle of induction, we have proved that the recurrence equation is correct.

Supplementary Explanation

Answer. The recurrence $T(1) = 1$, $T(n) = 2T(n-1) + 1$ counts all allowed committees for every $n \geq 1$.

Derivation/proof. Base case: for one person, $\{1\}$ is the only nonempty committee, so $T(1) = 1$. Inductive hypothesis: $T(n-1)$ correctly counts nonempty committees on the first $n-1$ people. At size n, committees excluding n are counted by $T(n-1)$. Those containing n and at least one old member correspond bijectively to a nonempty old committee and give another $T(n-1)$. The sole remaining committee is $\{n\}$. The disjoint cases exhaust all possibilities, so $2T(n-1) + 1$ is correct. Induction completes the proof.

Removing the new person gives an old nonempty committee whenever the original committee also contains an old member; adding that person back is the inverse map.

I068

Recursion

2015 Final | Question 9 | Source page 15 | Difficulty 3 | 5 marks
Q-2015-FINAL-P09-closed-form | Answer provenance: official

Let A and B be two positive integers represented as binary sequences, each n bits long. Assume n is a power of 2. The product $A \cdot B$ can be computed recursively as follows:

1. If $n = 1$, calculate $A \cdot B$ directly.
2. If $n > 1$, partition the bit representation of each number into two parts, the high $n/2$ bits and low $n/2$ bits. Calculate $A \cdot B$ using the following formula:

$$\begin{aligned} A \cdot B &= (A_h \cdot \underbrace{10\cdots0}_{n/2} + A_l) \cdot (B_h \cdot \underbrace{10\cdots0}_{n/2} + B_l) \\ &= (A_h \cdot B_h) \cdot \underbrace{10\cdots0}_{n} + (A_h \cdot B_l) \cdot \underbrace{10\cdots0}_{n/2} \\ &\quad + (A_l \cdot B_h) \cdot \underbrace{10\cdots0}_{n/2} + (A_l \cdot B_l). \end{aligned} \tag{2}$$

where A_h and B_h denote the high $n/2$ bits of A and B, respectively, and A_l and B_l denote the low $n/2$ bits of A and B, respectively. The terms in ( ) are calculated recursively.

Suppose the calculations take 1 unit of time when $n = 1$, and suppose that calculation of the expression (2) takes cn units of time given the results of the recursive calls, where c is a constant.

Let $T(n)$ be the total time it takes to compute $A \cdot B$. Write a recurrence for $T(n)$, find a closed-form expression for it, and simplify the closed-form expression using the big Θ notation.

find a closed-form expression for it,

Official Answer

Source: PDF physical page 16 (2015-final.pdf).

According to the above expression of $A \cdot B$,

$$T(n) = 4T(\frac{n}{2}) + cn$$

Iterating the recurrence, we get:

$$\begin{aligned} T(n) &= T(2^j) \\ &= 4T(2^{j-1}) + 2^j c \\ &= 4(4T(2^{j-2}) + 2^{j-1}c) + 2^j c \\ &= 4^2 T(2^{j-2}) + 2 \cdot 2^j c + 2^j c \\ &= 4^2(4T(2^{j-3}) + 2^{j-2}c) + 2 \cdot 2^j c + 2^j c \\ &= 4^3 T(2^{j-3}) + 2^2 \cdot 2^j c + 2 \cdot 2^j c + 2^j c \\ &\ \ \vdots \\ &= 4^j T(1) + 2^{j-1} \cdot 2^j c + \ldots + 2 \cdot 2^j c + 2^j c \\ &= 4^j + 2^j c(2^j - 1) \\ &= (1+c)n^2 - cn \\ &= \Theta(n^2) \end{aligned}$$

Supplementary Explanation

Answer. $T(n) = (1+c)n^2 - cn$.

Derivation/proof. For $j = \log_2 n$, iteration gives $T(n) = 4^j T(1) + cn \sum_{i=0}^{j-1} 2^i = n^2 + cn(n-1) = (1+c)n^2 - cn$. Substitution into $4T(n/2) + cn$ reproduces the expression, and at $n = 1$ it equals 1.

I069

Recursion

2014 Final | Question 8 | Source page 12 | Difficulty 3 | 10 marks
Q-2014-FINAL-P08 | Answer provenance: derived

> Let a be a real number such that $a > 1$. Consider a function $T(n)$ over integers defined as follows
>
> $$T(0) = 1, \qquad T(n) = aT(n-1) + n, \quad \text{for } n > 0.$$
>
> Prove that $T(n) = \Theta(a^n)$.
>
> You might need the fact that, for any $x \neq 1$,
>
> $$\sum_{i=1}^{n} ix^i = \frac{nx^{n+2} - (n+1)x^{n+1} + x}{(1-x)^2}.$$

Official Answer

Source: PDF physical page 12 (2014-final.pdf).

By iterating the recurrence, we get

$$\begin{aligned} T(n) &= aT(n-1) + n \\ &= a^2T(n-2) + a(n-1) + n \\ &= a^3T(n-3) + a^2(n-2) + a(n-1) + n \\ &\quad \cdots \\ &= a^n + \sum_{i=1}^{n} a^{n-i}i \end{aligned}$$

Hence, we have

$$\begin{aligned} T(n) &= a^n + \sum_{i=1}^{n} a^{n-i}i \\ &= a^n + a^n \sum_{i=1}^{n} a^{-i}i \\ &= a^n + a^n \frac{n(\frac{1}{a})^{n+2} - (n+1)(\frac{1}{a})^{n+1} + (\frac{1}{a})}{(1-a)^2} \\ &= \Theta(a^n). \end{aligned}$$

Supplementary Explanation

Answer. $T(n) = \Theta(a^n)$.

Derivation/proof. Iterating yields $T(n) = a^n + \sum_{i=1}^{n} a^{n-i}i = a^n\left(1 + \sum_{i=1}^{n} i(1/a)^i\right)$. Put $x = 1/a \in (0,1)$. The supplied finite-sum formula gives $\sum_{i=1}^{n} ix^i = [x - x^{n+1}(n+1-nx)]/(1-x)^2 \le x/(1-x)^2$, since $n + 1 - nx > 0$. All its summands are nonnegative, so

$$a^n \le T(n) \le \left(1 + \frac{a}{(a-1)^2}\right) a^n.$$

The two multipliers are positive constants independent of n, proving the tight bound.

In compact notation, the upper bound is $T(n) \le [1 + a/(a-1)^2]a^n$.

I070

Recursion

2014 Final | Question 10 | Source page 14 | Difficulty 3
Q-2014-FINAL-P10-closed-form | Answer provenance: official

> A computer system considers a string of digits to be a valid code word if and only if it contains an even number of 0's. For instance, 1203045 is valid, whereas 780900 is not. Let V_n be the number of valid code words of length n.
>
> It is clear that $V_1 = 9$ because "1", "2", ..., "9" are valid code words of length 1, while "0" is an invalid code word of length 1.
>
> Solve the recurrence to get a closed-form formula for V_n.

Official Answer

Source: PDF physical page 14 (2014-final.pdf).

A valid length n code can be obtained

- From a valid length $n-1$ code by appending "1", "2", . . ., or "9"to the end, or
- From an invalid length $n-1$ code by appending "0" to the end.

The total number of length $n-1$ codes is 10^{n-1}. Hence the number of invalid length $n-1$ codes is $(10^{n-1} - V_{n-1})$. So, we have

$$V_n = 9V_{n-1} + (10^{n-1} - V_{n-1}) = 8V_{n-1} + 10^{n-1}.$$

Iterating the recurrence, we get

$$\begin{aligned}
V_n &= 8V_{n-1} + 10^{n-1} \\
&= 8(8V_{n-2} + 10^{n-2}) + 10^{n-1} \\
&= 8^2 V_{n-2} + 8 \times 10^{n-2} + 10^{n-1} \\
&= \dots \\
&= 8^{n-1}V_1 + 8^{n-2} \times 10 \dots + 8 \times 10^{n-2} + 10^{n-1} \\
&= 9 \times 8^{n-1} + 8^{n-2} \times 10 \times (1 + \frac{10}{8} + \dots + (\frac{10}{8})^{n-2} + (\frac{10}{8})^{n-2}) \\
&= 9 \times 8^{n-1} + 8^{n-2} \times 10 \times \frac{(\frac{10}{8})^{n-1} - 1}{\frac{10}{8} - 1}) \\
&= 9 \times 8^{n-1} + \frac{10^n - 10 \times 8^{n-1}}{2} \\
&= \frac{1}{2}10^n + \frac{1}{2}8^n.
\end{aligned}$$

Supplementary Explanation

Answer. $V_n = (10^n + 8^n)/2$.

Derivation/proof. Iterating $V_n = 8V_{n-1} + 10^{n-1}$ gives $V_n = 9 \cdot 8^{n-1} + 10 \cdot 8^{n-2} \sum_{j=0}^{n-2} (10/8)^j = 9 \cdot 8^{n-1} + (10^n - 10 \cdot 8^{n-1})/2 = (10^n + 8^n)/2$.

For $n = 1$, $(10 + 8)/2 = 9$. Substituting into the recurrence gives $8(10^{n-1} + 8^{n-1})/2 + 10^{n-1} = (10^n + 8^n)/2$, independently checking the formula.

I071

Recursion

2014 Final | Question 10 | Source page 14 | Difficulty 3
Q-2014-FINAL-P10-recurrence | Answer provenance: official

> A computer system considers a string of digits to be a valid code word if and only if it contains an even number of 0's. For instance, 1203045 is valid, whereas 780900 is not. Let V_n be the number of valid code words of length n.
>
> It is clear that $V_1 = 9$ because "1", "2", ..., "9" are valid code words of length 1, while "0" is an invalid code word of length 1.
>
> For $n > 1$, find a recurrence for V_n by determining how it is related to V_{n-1}.

Official Answer

Source: PDF physical page 14 (2014-final.pdf).

A valid length n code can be obtained

- From a valid length $n-1$ code by appending "1", "2", . . ., or "9"to the end, or
- From an invalid length $n-1$ code by appending "0" to the end.

The total number of length $n-1$ codes is 10^{n-1}. Hence the number of invalid length $n-1$ codes is $(10^{n-1} - V_{n-1})$. So, we have

$$V_n = 9V_{n-1} + (10^{n-1} - V_{n-1}) = 8V_{n-1} + 10^{n-1}.$$

Supplementary Explanation

Answer. $V_1 = 9$ and $V_n = 8V_{n-1} + 10^{n-1}$ for $n > 1$.

Derivation/proof. Append a nonzero digit (nine choices) to each valid length-$n-1$ string, or append 0 to each invalid one. There are $10^{n-1} - V_{n-1}$ invalid strings, so $V_n = 9V_{n-1} + 10^{n-1} - V_{n-1}$.

The classes are disjoint and exhaustive: nine nonzero endings contribute $9V_{n-1}$ and a zero ending contributes $10^{n-1} - V_{n-1}$. The initial value is $V_1 = 9$.

I072

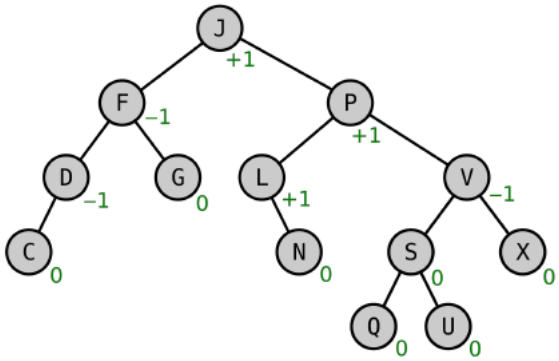
Recursion

2025 Spring Midterm | Question 5 | Source page 7 | Difficulty 4 | 12 marks
Q-2025S-MID-P05 | Answer provenance: derived

A height-balanced tree (HBT) is defined recursively as follows.

– A single node r is an HBT with height 0 and size 1. – Two nodes u, v form an HBT, where u is the root, and v is either u's left or right child. This HBT has height 1 and size 2. – Let T_1 and T_2 be two disjoint HBTs with roots r_1 and r_2, heights $h(T_1)$ and $h(T_2)$, sizes $n(T_1)$ and $n(T_2)$, respectively, such that $|h(T_1) - h(T_2)| \leq 1$. Create a new node r and make r_1 and r_2 the left and right child of r, respectively. The resulting structure is also an HBT with height $\max(h(T_1), h(T_2)) + 1$ and size $n(T_1) + n(T_2) + 1$.

The figure below shows an HBT T with height $h(T) = 4$ and size $n(T) = 13$. The numbers are the height differences between the left and right subtrees.



Show that $n(T) \geq f(h(T) + 2) - 1$ for any HBT T, where $f(k)$ is the k-th Fibonacci number. Recall that $f(1) = f(2) = 1$, and $f(k) = f(k-1) + f(k-2)$ for $k \geq 3$.

Official Answer

Source: physical PDF page 7 (COMP2711_S2025_Midterm.pdf), physical PDF page 8 (COMP2711_S2025_Midterm.pdf).

Base case: For HBT T which is simply a single node r, $h(T) = 0$ and $n(T) = 1$. $f(h(T) + 2) - 1 = f(2) - 1 = 0 \leq n(T)$.

Inductive step: Assume the proposition holds for some disjoint HBTs T_1 and T_2 with $|h(T_1) - h(T_2)| \leq 1$, i.e., $f(h(T_1) + 2) - 1 \leq n(T_1)$ and $f(h(T_2) + 2) - 1 \leq n(T_2)$.

Consider HBT T rooted at r with left subtree T_1 and right subtree T_2. By definition, $h(T) = \max(h(T_1), h(T_2)) + 1$ and $n(T) = n(T_1) + n(T_2) + 1$.

Case 1: $h(T_1) = h(T_2) + 1$. Then $h(T) = h(T_1) + 1 = h(T_2) + 2$ and

$$\begin{aligned} f(h(T) + 2) - 1 &= f(h(T) + 1) + f(h(T)) - 1 \\ &= f(h(T_1) + 2) + f(h(T_2) + 2) - 1 \\ &= (f(h(T_1) + 2) - 1) + (f(h(T_2) + 2) - 1) + 1 \\ &\leq n(T_1) + n(T_2) + 1 = n(T). \end{aligned}$$

Case 2: $h(T_2) = h(T_1) + 1$. The proof is analogous to that for Case 1.

Case 3: $h(T_1) = h(T_2)$. Then $h(T) = h(T_1) + 1 = h(T_2) + 1 \geq 1$. Note that $f(h(T) + 1) = f(h(T)) + f(h(T) - 1) \geq f(h(T)) + 0 = f(h(T))$, since Fibonacchi numbers are non-negative.

$$\begin{aligned} f(h(T) + 2) - 1 &= f(h(T) + 1) + f(h(T)) - 1 \\ &\leq f(h(T) + 1) + f(h(T) + 1) - 1 \\ &= (f(h(T_1) + 2) - 1) + (f(h(T_2) + 2) - 1) + 1 \\ &\leq n(T_1) + n(T_2) + 1 = n(T). \end{aligned}$$

Hence, by structural induction, $n(T) \geq f(h(T) + 2) - 1$ for any HBT T.

Supplementary Explanation

Answer. Every HBT generated by the definition satisfies $n(T) \geq f(h(T) + 2) - 1$.

Derivation/proof. For the single-node constructor, $h = 0, n = 1$ and $f(2) - 1 = 0 \leq 1$. For the two-node constructor, $h = 1, n = 2$ and $f(3) - 1 = 1 \leq 2$, whether the child is left or right.

Assume both subtrees satisfy the induction hypothesis and let $H = \max(h_1, h_2)$. If their heights differ, interchange left and right if necessary so that $h_1 = H, h_2 = H - 1$, with $H \geq 1$. The merged tree has height $H + 1$ and

$$n = n_1 + n_2 + 1 \geq f(H + 2) + f(H + 1) - 1 = f(H + 3) - 1.$$

If both heights equal $H \geq 0$, then $n \geq 2f(H + 2) - 1$. The initial Fibonacci values and nonnegative recurrence imply $f(H + 2) \geq f(H + 1)$; at $H = 0$ both values are one. Thus $2f(H + 2) - 1 \geq f(H + 2) + f(H + 1) - 1 = f(H + 3) - 1$. Both merge cases satisfy the bound, completing structural induction without using the undefined $f(0)$.

I073

Recursion

2021 Final Part 1 | Question 5 | Source page 3 | Difficulty 4 | 6 marks
Q-2021-FINAL1-P05 | Answer provenance: derived

> Give a recursive definition of the set of bit strings that have more 0s than 1s.

Official Answer

Source: physical PDF page 3 (2021-final-part-1.pdf).

> Let S be the set of bit strings that have more 0s than 1s.
>
> **Base case:** $0 \in S$.
>
> **Recursive case:** If x and y are bit strings in S, then $xy \in S$, $xy1 \in S$, $x1y \in S$ and $1xy \in S$.

Supplementary Explanation

Answer. Base: $0 \in S$. Rules: from $w = uv \in S$, insert 0, 01, or 10 at the split: $u0v, u01v, u10v \in S$. No other strings belong to S.

Derivation/proof. Soundness: the base has zero-minus-one balance 1. Inserting 0 increases this balance by one; inserting 01 or 10 preserves it. Therefore every generated string has more zeros. Completeness: use strong induction on length of a positive-balance string w. If w contains only zeros, construct it by repeated insertion of 0. Otherwise w contains both symbols and hence an adjacent unequal pair 01 or 10 at a transition. Delete that pair to obtain a shorter positive-balance string, which is generated by induction; reinsert the pair at its original split. This generates every required string, including all positions of the minority ones.

The split $w = uv$ may be at either endpoint. After deleting a mixed adjacent pair, the balance remains positive, so the shorter string is nonempty. Reinsertion at the original split recovers the string even when its ones occur internally. The independently derived insertion definition is retained as an alternative to the printed two-string construction, not passed off as its wording.

In the completeness step, deleting that pair preserves the balance, while the length decreases by exactly two. The induction hypothesis supplies the shorter string, and we reinsert it at the original split.

I074

Recursion

2021 Final Part 2 | Question Bonus(b) | Source page 4 | Difficulty 4 | 8 marks
Q-2021-FINAL2-BONUS-b | Answer provenance: official

> Let G be an undirected complete graph having n vertices. For a vertex v, $N(v)$ denotes the set of neighbors of v in G. A random walk on G is the following process, which occurs in a sequence of discrete steps: starting at a vertex v_0, we proceed at the first step to a neighbor of v_0 chosen uniformly at random. This may be thought of as choosing a random edge incident on v_0 and walking along it to a vertex $v_1 \in N(v_0)$. At the second step, we proceed to a randomly chosen neighbor of v_1, and so on.
>
> Redo (a) for the case where G is a line graph of n vertices, and u is at one end and v is at the other end.

Official Answer

Source: physical PDF page 5 (2021-final-part-2.pdf).

> (b) Number of the vertices as $1, \ldots, n$. Let T_i denote the number of steps to go from i to n. We have $T_n = 0$, so $E[T_n] = 0$. The goal is to find $E[T_1]$.
>
> From vertex 1, we can only go right, so $T_1 = 1 + T_2$ and $E[T_1] = 1 + E[T_2]$. Rewrite it as $E[T_1] - E[T_2] = 1$. For any $1 < i < n$, we go left or right with equal probability, so $E[T_i] = 1 + \frac{1}{2}E[T_{i-1}] + \frac{1}{2}E[T_{i+1}]$. Rearrange it as $E[T_i] - E[T_{i+1}] = 2 + E[T_{i-1}] - E[T_i]$. Now we have
>
> $$\begin{aligned} E[T_1] - E[T_2] &= 1 \\ E[T_2] - E[T_3] &= 3 \\ &\cdots \\ E[T_{n-1}] - E[T_n] &= 2n - 3. \end{aligned}$$
>
> Summing up, we have $E[T_1] = E[T_n] + 1 + 3 + \cdots + 2n - 3 = (n-1)^2$.

Supplementary Explanation

Answer. $(n-1)^2$.

Derivation/proof. Label the path vertices 1,...,n and let h_i be expected steps to reach n. Then h_n=0 and h_1=1+h_2. For 1<i<n, h_i=1+(h_(i−1)+h_(i+1))/2. Define d_i=h_i−h_(i+1); rearrangement gives d_1=1 and d_i=d_(i−1)+2, hence d_i=2i−1. Telescoping yields h_1−h_n=1+3+...+(2n−3)=(n−1)². Expectations are finite: from any vertex there is a positive uniformly bounded chance to follow right steps to n within n−1 steps, giving a geometric block tail. For n=1 the answer is zero.

For $n \geq 2$, from any non-target vertex, following rightward steps reaches the target within $n-1$ steps with conditional probability at least $2^{-(n-1)} > 0$. Each block has this uniform success lower bound, so a geometric tail controls the hitting time. Explicitly,

$$h_1 = h_1 - h_n = \sum_{i=1}^{n-1} d_i = 1 + 3 + \cdots + (2n-3) = (n-1)^2.$$

For $n = 1$, both endpoints coincide and the expectation is zero.

The difference recurrence is $d_i = d_{i-1} + 2$, giving $d_i = 2i - 1$.

I075

Recursion

2021 Final Part 2 | Question 1 | Source page 1 | Difficulty 4 | 10 marks
Q-2021-FINAL2-P01 | Answer provenance: official

> Recall the definition of Fibonacci number:
>
> $F(1) = 1, F(2) = 1, F(n) = F(n-1) + F(n-2)$ for $n \geq 2$.
>
> Give a combinatorial proof to the following identity:
>
> $$F(n+2) = \sum_{i=0}^{(n+1)/2} \binom{n+1-i}{i}.$$
>
> You can assume $n \geq 1$ and n is odd.
>
> [Hint: Consider bit strings with no consecutive 0s.]

Official Answer

Source: physical PDF page 1 (2021-final-part-2.pdf).

> Let S_n be the set of bit strings of length n with no consecutive 0s. We will show that both sides are equal to $|S_n|$. Let s_i denote any bit string in S_i.
>
> Left-hand side: For $n = 0$, the empty string is the only string and it has no consecutive 0s. So, $F(0 + 2) = 1 = |S_0|$ bit string that have no consecutive 0s. For $n = 1$, there are $F(1 + 2) = F(3) = F(2) + F(1) = 2 = |S_1|$ bit strings of length 1 that have no consecutive 0s. For $n > 1$, a bit string of length n that has no consecutive 0s can be obtained by appending a "1" to s_{n-1} and appending a "10" to s_{n-2}. The two strings must be different and these two are the only possible cases. By induction, we have $|S_n| = F(n + 2)$.
>
> Right-hand side: $\binom{n+1-i}{i}$ is counting the number of bit strings of length n that has exactly i 0s and no consecutive 0s. When a bit string of length n that has more than $(n + 1)/2$ 0s, there must be at least two consecutive 0s. So, the expression sums over all mutually disjoint cases, and thus it equals to $|S_n|$.

Supplementary Explanation

Answer. Both sides count length-n bit strings without consecutive zeros.

Derivation/proof. Let s_n count these strings. At length zero s0=1 and length one s1=2. For n≥2, a valid string ends either in 1 after a valid length-(n−1) prefix, or in 10 after a valid length-(n−2) prefix; these disjoint exhaustive cases give s_n=s_(n−1)+s_(n−2). Hence s_n=F(n+2). Independently fix i zeros. Arrange n−i ones; their n−i+1 gaps (including endpoints) can each contain at most one zero, giving C(n−i+1,i). Feasibility requires i≤(n+1)/2; summing disjoint zero counts proves the formula. The forward recurrence only needs Fibonacci indices ≥3; the printed boundary n≥2 is not used to evaluate an unspecified F(0).

The gap condition is $i \leq n - i + 1$, equivalent to $i \leq (n + 1)/2$. The stipulated odd n makes the upper limit integral. Both decompositions are unique and the final-block cases are disjoint and exhaustive.

The recurrence is $|S_n| = |S_{n-1}| + |S_{n-2}|$, and the fixed-zero count is $\binom{n-i+1}{i}$.

I076

Recursion

2018 Final | Question 3(b) | Source page 5 | Difficulty 4
Q-2018-FINAL-P03-b | Answer provenance: official

> S, a set of integers, is recursively defined as follows:
>
> Basis: 4914 and 3588 are in S.
>
> Recursive step: If $x, y \in S$, then $x - y \in S$.
>
> (Note that x and y may be the same.)
>
> (b) Let A_m be the set of all multiples of some integer m. Find the integer m such that $S = A_m$, and provide a complete proof of this equality, showing all steps.

Official Answer

Source: physical PDF page 5 (2018-final.pdf).

Part (a) shows that $4914s \in S$ and $3588t \in S$ for all s, t, so $\gcd(4914, 3588) \in S$. Therefore

$$\begin{aligned}\gcd(4914, 3588) &= \gcd(3588, 1326) = \gcd(1326, 936) = \gcd(936, 390) \\ &= \gcd(390, 156) = \gcd(156, 78) = 78 \in S.\end{aligned}$$

$A_{78} \subseteq S$: Using part (a) directly, since $78 \in S$, then $78n \in S$ for any integer n. $S \subseteq A_{78}$: 4914 and 3588 are multiples of 78. Suppose x and y are multiples of 78. Then clearly $x - y$ is a multiple of 78. By structural induction, $S \subseteq A_{78}$.

Supplementary Explanation

Answer. $m = 78$ is the positive generator, and $S = 78\mathbb{Z}$ (also $A_{-78} = A_{78}$).

Derivation/proof. Euclid gives $4914 = 3588 + 1326$, $3588 = 2 \cdot 1326 + 936$, $1326 = 936 + 390$, $936 = 2 \cdot 390 + 156$, $390 = 2 \cdot 156 + 78$, $156 = 2 \cdot 78$. Every remainder belongs to S: previous entries and their multiples belong by part (a), and differences are allowed. Thus $78 \in S$, and part (a), together with $0 \in S$, gives $78\mathbb{Z} \subseteq S$. Conversely $4914 = 63 \cdot 78$ and $3588 = 46 \cdot 78$. Differences of multiples of 78 remain multiples, so structural induction gives $S \subseteq 78\mathbb{Z}$. Both inclusions prove equality.

The basis numbers satisfy $4914 = 63 \cdot 78$ and $3588 = 46 \cdot 78$. A generator $m = -78$ gives the same set as $m = 78$.

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I027	Q-2020-FINAL1-P02	1
I028	Q-2019-FINAL-BONUS-a	17
I029	Q-2019-FINAL-BONUS-b	17
I030	Q-2018-FINAL-P02-a-E1	4
I031	Q-2018-FINAL-P02-a-E2	4
I032	Q-2018-FINAL-P04-c-basis	6
I033	Q-2017-FINAL-P10-a-first	14
I034	Q-2017-FINAL-P11-a-T1	15
I035	Q-2025F-FINAL-P10-i	17
I036	Q-2024F-FINAL-P09-b-i	24
I037	Q-2024F-FINAL-P09-b-ii	24
I038	Q-2023F-FINAL-P08-b	12
I039	Q-2023F-FINAL-P09-a	14
I040	Q-2023F-FINAL-P09-b	14
I041	Q-2020-FINAL1-P05-a-formula	3
I042	Q-2019-FINAL-P06-a	8
I043	Q-2018-FINAL-P02-a-E3	4
I044	Q-2018-FINAL-P04-a	6
I045	Q-2018-FINAL-P04-b	6
I046	Q-2018-FINAL-P04-c-edges	6
I047	Q-2017-FINAL-P10-a-second	14
I048	Q-2017-FINAL-P10-b	14
I049	Q-2017-FINAL-P11-a-T2	15
I050	Q-2017-FINAL-P11-b	15
I051	Q-2016-FINAL-P07-a-recurrence	13

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I054	Q-2015-FINAL-P09-recurrence	15
I055	Q-2025F-FINAL-P10-ii	17
I056	Q-2024S-FINAL-P05	6
I057	Q-2023F-FINAL-P09-c	14
I058	Q-2023F-FINAL-P09-d	14
I059	Q-2021-FINAL1-P02	1
I060	Q-2021-FINAL1-P06	3
I061	Q-2020-FINAL1-P05-b	3
I062	Q-2020-FINAL1-P06-b	5
I063	Q-2019-FINAL-P06-b	8
I064	Q-2018-FINAL-P02-b	4
I065	Q-2017-FINAL-P11-c	15
I066	Q-2017-FINAL-P11-d	15
I067	Q-2016-FINAL-P07-a-proof	13
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I069	Q-2014-FINAL-P08	12
I070	Q-2014-FINAL-P10-closed-form	14
I071	Q-2014-FINAL-P10-recurrence	14
I072	Q-2025S-MID-P05	7
I073	Q-2021-FINAL1-P05	3
I074	Q-2021-FINAL2-BONUS-b	4
I075	Q-2021-FINAL2-P01	1
I076	Q-2018-FINAL-P03-b	5